dune-rover report

Predicting slip, sinkage and traversability of wheeled robots on dry sand and dunes: a literature review, a gap analysis for Arabian dunes, and a minimal reproducible experiment plan

Author: Huda Mueen. Repository: `hudasol/dune-rover`. Date of all retrievals: 2026-10-11. Related work, not repeated here: `hudasol/dhill` (heat-and-dust operating envelope). This project is about terrain mobility only.

> **Read this first.** This is a research document. It is not engineering approval, a safety case or a performance claim for any real vehicle. Three kinds of content appear and are always labelled: **PUBLISHED**: a number read in a retrieved source (every one is a row in the evidence table, Appendix A, with URL, passage and retrieval date). **SIMULATED-BY-ME**: output of code in this repository (section 8). It depends on stated model assumptions and is **never field evidence**. **PLANNED**: a protocol that has not been run (section 9).

---

0. Executive summary

Question. How well can slip, sinkage and traversability of wheeled (secondarily legged) robots on dry sand and dunes be predicted from terrain data and models; how well does that transfer from lab to field; and what would a minimal open-data planner for Arabian dunes need that the literature does not provide?

Method. A written protocol (`docs/PROTOCOL.md`) came first. Five row-extraction passes (terramechanics, simulation, learning, field and analog, legged and wheel design) collected 144 evidence rows from retrieved sources, and a sixth pass audited open data, dune morphology and UAE facts. Separate agents then re-opened the sources for **every one of the 144 rows** (30 randomly drawn plus 7 flagged in phase 1, the other 107 in phase 2). A simulation-only slope-and-slip traversability study was run on real Copernicus GLO-30 elevation data from the Liwa and Rub' al Khali areas,

with one-at-a-time and global (Latin hypercube) sensitivity analysis. Phase 2 added a sub-grid slip face detector, a three class map (TRAVERSABLE, BLOCKED, UNKNOWN) with Monte Carlo over the four assumed parameters, a second morphology (linear dunes), and a pre-registered field kit.

Headline findings (published evidence).

1. **Classical and granular models work only after per-soil fitting, and mostly on level ground.** Bekker-Wong style drawbar-pull errors are about 24 to 27 % (median) on two lunar-simulant wheel datasets and sinkage errors 28 to 52 %, but reach 135 % and 385 % on a firm soil with a small light wheel (SIM-025 to SIM-027). Granular RFT gives drawbar mean absolute errors of roughly 5 to 7 N on medium sand and 15 N on a fine simulant (TM-001 to TM-004). The Janosi shear modulus K, which controls slip, differs about 75-fold between papers (0.0006 to 0.045 m, the top of the range from poppy seeds), so it is tuned rather than transferred.
2. **Speed and slope break the quasi-static assumptions.** On olivine sand the drawbar coefficient fell 8 % between 0.01 and 0.2 m/s and 63.7 % between 0.2 and 1 m/s (TM-018, TM-019). A rover-scale Bekker-Wong slope model had untuned side-slip errors of 14.8 % at 10 degrees and 25.9 % at 20 degrees on Toyoura sand (TM-022, TM-023, a thesis); the 0.49 % at 20 degrees came after tuning on that same case (TM-024, in-sample).
3. **Field slip on sand is large, patchy and not predictable from slope or appearance.** Spirit measured 15 to 20 % slip at 19 degrees and 95 % slip at 16 degrees about 30 m later on terrain that looked alike (FLD-011, FLD-012). Opportunity's visual-odometry slip stayed at 98.9 to 99.5 % at Purgatory ripple (FLD-002). Perseverance exceeded 94 % at about 20 degrees (FLD-025). Visual odometry, the field slip sensor, itself degrades on bland sand (FLD-041).
4. **Learned predictors are validated mostly in-distribution.** The strongest held-out result is a Perseverance model (AUROC 0.874 on held-out sols, LRN-010) on Mars. No retrieved paper holds out terrain type (train without sand, test on sand), and none reports a learned slip or sinkage error on Earth dunes.
5. **Arabian evidence is almost empty.** The only Arabian wheeled-robot field evidence found is qualitative: a rover in Dhofar, Oman, met looser sand than expected and got stuck (FLD-042). No Bekker or RFT parameters for Arabian dune sand, no grain-size map of the Empty Quarter and no open LiDAR or drone elevation model of its dunes were found. The UAE-authored wheel work found is single-wheel lunar-simulant testing (not sand).

6. **Open elevation data resolve dunes but not slip faces.** Copernicus GLO-30 (about 28 by 31 m posts at 23.5 N, free licence with attribution) resolves dune crests and interdune corridors for dunes tens to hundreds of metres high. Slip faces of the large Rub' al Khali dunes are only 1 to 4 posts wide, so slope read from the grid is biased low; 90 m products remove 80 to 100 % of the cells steeper than 25 degrees in the three windows used here.

Headline findings (SIMULATED-BY-ME, model-assumption dependent, not field evidence). With a stated friction-circle slip model and nominal assumptions, the 30 m grid blocks only 0.2 to 1 % of travel directions in three real dune windows, because few cells exceed the model's nominal climbing limit of 25.5 degrees. That is mostly a statement about grid resolution and assumptions, not evidence that the dunes are easy. Across the plausible assumption ranges the model's climbing limit varies from 7.6 to 34.7 degrees (5 to 95 %: about 15 to 32 degrees). The assumptions that change the map most are the friction angle, the form of the slip law, the DEM posting, the slip scale i0 and the motion-resistance coefficient. DEM noise and the planning slip limit matter least (section 8).

Headline findings, phase 2 (SIMULATED-BY-ME, model-assumption dependent; detector validated on synthetic truth only). A 30 m DEM cannot see slip faces, so each cell now carries a modelled probability that an unresolved active face lies inside it and the map has three classes. At 30 m the **UNKNOWN share is 8.6 % in A south Liwa, 10.9 % in B (megadunes) and 2.5 % in C (Saudi Rub' al Khali linear dunes)**; BLOCKED is under 0.1 % everywhere, because the DEM hardly shows slopes the model calls impassable. If a face within one post of the cell counts (a post hoc variant) the UNKNOWN share is 34.4 %, 32.2 % and 22.0 %. 99.1 % (A), 97.0 % (B) and 100.0 % (C) of cells have the same slope based verdict under all 256 draws of the four assumed slip parameters, so the three class map is far more sensitive to the sub-grid morphology assumptions and the resolution than to those four parameters (section 8.7). Of 18 pre-registered predictions, the hits and misses are in `docs/PREDICTIONS.md`; most of the quantitative ones about the simulation missed.

Verification. All 144 rows now have an independent re-check against the source (section 11). Material errors: **6 of 144 (4.2 %, Wilson 95 % interval 1.9 to 8.8 %)**, all in the 107 rows checked in phase 2 (6 of 107, 5.6 %); any discrepancy including minor label errors: 36 of 144 (25.0 %). All are corrected in the table. The phase 1 random sample (0 of 30 material) had understated the rate. The unit of the slope term in SIM-013 is not stated in its source, so that row is flagged and must not be used numerically outside the tested envelope.

Biggest gap and first experiment. There is no published slip-versus-slope dataset for a wheeled robot on dune sand with full soil characterisation. The first field experiment to run is the **slope ladder** (section 9): one small rover with fixed wheels, one dune stoss slope, runs at about 0, 5, 10, 15, 20 and 25 degrees with RTK ground truth and full soil description, calibrate on the gentle slopes and predict the steep slopes blind.

---

1. Scope, protocol and method

Protocol. Written before searching, as the file itself states (`docs/PROTOCOL.md`; no independent timestamp): search strategy by query family, inclusion criteria, comparability rules, extraction fields, a verification plan and a stopping rule. One deviation is documented: the verification pass was run on the draft evidence table, before this report's prose was written (an ordering from my recollection of the workflow, not logged), so the report quotes corrected values. A second, independent pass on this report's own claims is described in section 11.

Evidence classes are kept visibly separate (see the banner). Published results also carry a validation type: `lab` (bench, soil bin, tilt bed), `field` (real vehicle on real terrain, including Mars), `simulation-only`, and `sim-vs-lab` or `sim-vs-field` where a physical comparison is reported. Of 144 rows: 66 lab, 38 field, 20 simulation-only, 18 sim-vs-lab, 1 sim-vs-field, 1 none (an unreferenced capability statement kept for the record). A physical test on a prepared bed, tilt platform or engineered analog site (for example the JPL Mars Yard) is classed `lab`; a real vehicle on natural terrain, including Mars, is `field`.

Inclusion. A result became a row only if the number was found in retrieved text (table, caption or body), the terrain is dry granular or sand-like (or flagged as a comparison), at least one mobility metric is reported, and vehicle and soil are described or marked "not reported". Paywalled sources not retrieved are listed in the per-topic notes under `research/raw/` with the exact URL to fetch; no number is taken from them.

Comparability. Rows are never averaged or ranked across setups. Each row has a `comparability_warning` where slip definition (wheel odometry, visual odometry, velocity-based, distance-based), slope definition (terrain versus pitch, or emulated by horizontal pull), gravity, wheel geometry, soil, speed or validation type differ.

Parameters. No soil parameter is invented. Each is either cited with its location or labelled ASSUMPTION with a sensitivity range (section 8, Table 8.1).

Quotation and licences. Sources are paraphrased. The licence of every dataset and library used is in `docs/DATA_AND_LICENSES.md` (summarised in section 5).

Limits of the search. The search tool was US-only and returned a limited number of hits per query. Arabic-language and regional literature, paywalled journals (for example the Journal of Terramechanics, Science) and several standard references (Bekker 1969, Wong's textbook, Wong and Reece 1967) were not retrieved; absence of a result is therefore a statement about this search, not about the world.

---

2. Background in brief (for newcomers; see `docs/GLOSSARY.md`)

A wheel on loose sand sinks, pushes sand aside (motion resistance) and shears the sand beneath it to push the vehicle forward (traction). **Slip** measures how much the wheel turns without advancing. **Sinkage** is the depth the wheel penetrates. **Drawbar pull** is the traction left over after resistance is paid. On a slope, part of the vehicle's weight must be paid for with drawbar pull, so slip rises with slope until the vehicle can no longer climb (entrapment). Dry sand has almost no cohesion and a friction angle of roughly 28 to 38 degrees in the lab papers retrieved; wind-blown dune slip faces sit near the angle of repose, 32 to 35 degrees (M05, Kroy et al.; `research/raw/morphology_rows.csv`).

The older family of models (Bekker; Wong and Reece; Janosi and Hanamoto) describes the soil with a pressure-sinkage law (constants kc, kphi, n) and a shear law (cohesion, friction angle, shear modulus K), then integrates stress under the wheel. A newer family (granular resistive force theory, RFT) sums depth- and orientation-dependent forces on small wheel elements. Discrete-element methods (DEM) simulate individual grains, and Project Chrono offers fast deformable-terrain (SCM) and continuum (CRM) alternatives. Learning methods predict slip or hazard from images, proprioception or both.

---

3. Evidence synthesis

Row identifiers refer to Appendix A. Every number below is **PUBLISHED** unless marked.

3.1 Classical Bekker-Wong style models

- **Parameters are soil-, geometry- and tuning-dependent.** On dry Tottori sand the fitted kphi roughly doubles (about 430 to 800-880) when plate aspect ratio drops from 6 to 2-1.4, and kc rises about four-fold (from

about 2.5 to 3, up to 9 to 10); sinkage goodness of fit ranged 65 to 95 % (TM-029 and its table). Shear moduli K found in the retrieved papers range from 0.0006 m to 0.045 m (75-fold; the top is poppy seeds), so K behaves as a tuning knob. The Wong textbook table of dry-sand constants was not retrieved; one secondary table found carries an apparent transcription error and is not used.

- **Error on benchmark wheels.** RoverDevKit's own evaluation of a Bekker-Wong wheel kernel against published single-wheel data reports median drawbar errors of 24 % (Rashid-1 style wheel on FJS-1 simulant) and 27 % (Ding wheel data), sinkage errors of 28 and 52 %, and 135 % (drawbar) and 385 % (sinkage) for a small light wheel on dense simulant (SIM-025 to SIM-027). These are the authors' evaluations of other authors' data, sim-vs-lab, not field.
- **Slope.** The only slope result is a thesis: a Bekker-Wong-Ishigami model for a 7 kg rover on Toyoura sand gave side-slip errors of 14.76 % at 10 degrees and 25.91 % at 20 degrees with parameters borrowed from another lab, and 0.49 % at 20 degrees after tuning on that case (TM-022 to TM-024). The error metric is a side-slip distance ratio, not slip ratio, and the sandbox is 0.08 m deep.
- **Rate dependence.** Measured drawbar coefficient on olivine sand fell 8 % from 0.01 to 0.2 m/s and 63.7 % from 0.2 to 1 m/s at 20 % slip (TM-018, TM-019), against the quasi-static assumption. Slip-rate effects were also reported for grousered wheels (TM-021; the 85 % "accuracy" there is an in-sample goodness-of-fit, per the verification).

3.2 Granular RFT and scaling

- RFT predicts rigid-wheel drawbar pull and sinkage from a small parameter set. On Quikrete medium sand the mean absolute drawbar error was 5.55 to 7.11 N across loads (TM-001, TM-002); on Mars Mojave Simulant (fines) 15.11 N and 18.41 mm sinkage at 190 N (TM-003, TM-004). Error grows with load; RFT under-predicts drawbar at low positive slip and over-predicts at |slip| above 0.4 (per the source's discussion). The response surfaces for both the medium sand and the fines were scaled from poppy-seed data rather than measured directly, so the medium-sand results carry that caveat too.
- On poppy seeds (not sand), RFT sinkage error was 2.73 mm against 26.07 mm for the Bekker-type model but drawbar error was larger for RFT (5.05 N versus 3.41 N) (TM-005, TM-006; the plate in the Bekker fit was far smaller than the wheel patch).
- Granular scaling laws held within about 3 % on level Quikrete sand for pairs of wheels (TM-007). Their extension to slopes and lunar gravity has been checked only in DEM (slip RMSE 0.015, TM-020, simulation-only). No physical sloped or low-gravity scaling validation was found.

- Legged and plate RFT follow-ups agree with controlled-box experiments within roughly 3 to 15 % (up to 30 % for reversed legs; TM-008, TM-009, TM-011, TM-012), two of the four on poppy seeds rather than sand and 10.5 and 11.5 % mean speed and height error against a beach (TM-010, sim-vs-field, legged). A humanoid trained with an RFT-based simulator walked on real dry sand at up to 2.5 m/s commanded (LEG-015).

3.3 DEM, multibody and Project Chrono

- **Accuracy where a number exists.** A 2D DEM of a grouser wheel on Toyoura sand gave 12.1 % slip against 14.7 +/- 3.0 % in the lab (SIM-018). A 36-million-particle DEM of a rigid tyre gave pull 15 % of weight against 17 % lab and gross traction 20 % against 31 % at 20 % slip, under-predicting sinkage and resistance (SIM-019). A soft foot in silica sand: mean pressure-sinkage difference 13.5 % (SIM-021).
- **Many validations are plots only.** Chrono CRM and DEM wheel and rover comparisons against lab data (NASA flat and tilted beds, SIM-001, SIM-002, SIM-009; the Rashid-1 wheel rig, SIM-015) are described as close agreement but give no scalar error. Slope-climbing limit error is not reported anywhere in the retrieved set.
- **Calibration is not independent evidence.** SCM parameters fitted to a DEM gave a negative Kc and a friction angle of about 21 degrees (a numerical fit, SIM-005); DEM used scaled particles and tuned friction (SIM-015, SIM-018).
- **Cost.** Wall-clock to simulated time ranges from about 9 million (fine-particle CPU DEM, SIM-020, derived) to hundreds or thousands (GPU DEM, SIM-016, SIM-024) to about 11 to 23 (Chrono CRM) and 18 to 51 (Chrono SCM) on one benchmark (SIM-011), and about 0.5 ms per closed-form Bekker-Wong wheel solve (SIM-028; no simulated-time ratio reported). The SCM-versus-CRM ordering depends on the benchmark.
- **Gravity.** Simulating a 440 kg rover at lunar gravity predicted far higher slip than a light 1 g comparison (about 85 % versus about 42 % at 30 degrees, SIM-003, where the source of the 42 % is not stated; about 75 % versus 18 % at 10 degrees, SIM-004), with no lunar validation (simulation-only).
- Project Chrono is BSD-3 licensed (verified from its LICENSE file). **It is not used by this repository's code.** No retrieved source compares Chrono, DEM or any multibody simulator with a full rover on real dune sand.

3.4 Learning-based slip and traversability

- **Data needs.** Visual slip prediction needed 1000 to 2000 stereo frames per terrain with slip labels from visual odometry (LRN-003, LRN-004). The Perseverance model used about 43,000 self-labelled images from

about 500 sols and 45 km (LRN-010). A proprioceptive slip classifier used about 13,000 samples (9235 train, 3958 test) from one single-wheel testbed (LRN-009).
- **Best held-out evidence.** AUROC 0.874, F1 0.758 on held-out Perseverance sols, including a sand dune patch at about 0.3 slip (LRN-010, LRN-011). Held out by sol, not by terrain type; Mars, not Earth sand. Removing proprioceptive alignment dropped AUROC to 0.723 (LRN-011): appearance alone does not fix mechanics.
- **Documented failures.** Test error on gravel rose to 27.5 % against 7.16 % in training because the test roll/pitch combinations were absent from training (LRN-003). Misclassifying soil or gravel as sand inflated whole-pipeline error from 11.2 to 21.8 % (LRN-004). The moderate-slip class was the weakest in proprioceptive classification, about 75 % against about 85 % for low and high slip (LRN-009), and that test was a random hold-out on one testbed.
- **Transfer.** A policy trained in simulation transferred zero-shot to a rover on simulant at 20 degrees: travel loss fell from 81.9 to 43.0 % on dry sand and from 100 % (entrapment) to 30 % on wet sand (LRN-019, sim-vs-lab, trial count not stated).
- **Datasets.** No public dataset with Earth dune or desert sand and measured wheel or foot slip labels was found. TAIL (sandy sites, SLAM benchmark, GPLv3, no slip labels) and the Perseverance driving set (Martian, release pending, no licence seen) are the closest.

3.5 Field and analog evidence

- **Mars rovers.** Spirit: 15 to 20 % slip on 19 degrees, 95 % on 16 degrees (FLD-011, FLD-012), up to 50 % at 10 to 13 degrees (FLD-010), over 100 % in places (FLD-009, FLD-013). Opportunity: about 50 m of wheel travel for 2 m of track at Purgatory ripple (FLD-001), slip 98.9 to 99.5 % (FLD-002), ripple 30 cm high, trapped several weeks (FLD-040). Curiosity: 3D slip above 70 % on shallow megaripples (FLD-018). Perseverance: above 94 % at about 20 degrees (FLD-025), slip limits above 50 % needed on a 15 to 20 degree hill (FLD-024).
- **Not a clean function of slope.** Segment-averaged Spirit slippage on sand-sized terrain rose from 4.0 % (0 to 5 degrees) to 9.9 % (5 to 10 degrees) (FLD-015, FLD-016), while fine-grain terrain approached 100 % at 10 to 15 degrees (FLD-014). The authors warn that steep segments were avoided or visual-odometry corrected, so higher slopes sometimes show less slippage (warning on FLD-015). Do not fit a clean slip-versus-slope line to these aggregates.
- **Lab-to-field.** Engineering slip tables built from a dry-sand tilt bed worked on pure sand but failed on mixed or hidden weak deposits (Maimone et al.); separately, Lamarre and Kelly report that vision-based slip prediction fails on sand (both summarised in the field notes). The tilt-

bed figures that are quantitative (90 % slip at 20 degrees, 25 degrees climbing limit, FLD-006, FLD-007) come from a press release, so FLD-006 is excluded from the anchor comparison in section 8.

- **Tilt beds on Earth.** A smooth-wheel 35 kg rover (El-Dorado-II) showed slip close to 0.9 at 4 degrees without grousers and about 0.2 with grousers (FLD-030, FLD-031); a 23 kg rover with 296 mm wheels made less than 10 mm progress per revolution at 30 degrees, the angle of repose, whatever the grouser length (FLD-035); a 400 kg Scarab exceeded 90 % at 20 degrees and had no uphill progress at 25 degrees (FLD-027, FLD-028).
- **Arabian and Gulf.** One qualitative report (FLD-042): the AMADEE-18 Husky rover in Dhofar, Oman, met much looser and softer sand than expected. Nothing with slip, sinkage or slope numbers was found for Rub' al Khali, Saudi, Qatari, Kuwaiti or Emirati dunes. The UAE-authored single-wheel rig results (slip 0 to 0.75, sinkage about 1 to 2 cm) use lunar simulant and are not evidence for desert sand. See also `research/raw/field_notes.md` for non-Arabian desert trials, which were qualitative.

3.6 Legged and tracked comparisons (secondary)

- **Legs.** Gait dominates: the same SandBot ran about 2 cm/s with a hard-ground gait and 8 cm/s with a soft-ground gait (LEG-001, LEG-002), and about 20 cm/s versus 2 cm/s for a 5 % change in packing (LEG-003, LEG-004). On slopes a 350 g hexapod lost about two thirds of its speed between 0 and 20 to 24 degrees (LEG-007 to LEG-009; the same row set appears as FLD-037, FLD-038). On a 22 kg quadruped (SpaceBok) planar feet slipped near the soil friction angle while point feet sank up to 120 mm (LEG-010, LEG-012, LEG-013; testbed limit 25 degrees). All lab beds, none dune sand.
- **Wheel geometry.** Larger and wider wheels cut slip strongly on Toyoura sand: slip 0.741 at D = 116 mm versus 0.056 at D = 327 mm at an equivalent 12 degrees (LEG-017, LEG-018), and 0.471 at narrow versus 0.247 at wide at 14 degrees (LEG-019, LEG-020), with slope emulated by horizontal pull and from slides, not a paper. Grouser benefit saturates (LEG-021). A flexible metallic wheel gave more than twice the drawbar pull of a rigid one at 1 m/s on olivine sand (LEG-022).
- **Tracks.** The one direct wheel-versus-track comparison found could not be attributed unambiguously and is not entered as a row. No extractable numeric tracked-vehicle sand data were retrieved. No same-vehicle wheel-versus-leg-versus-track comparison on dry sand was found.

---

4. Taxonomy of prediction approaches and when each is appropriate

The table is a judgement made from the evidence above, not a ranking from head-to-head tests. No retrieved study compares these families on the same vehicle and the same dune sand.

Family	Needs	Strengths (published)	Weaknesses (published)	Appropria when
Slope-threshold or friction-circle rules (the model in section 8)	Slope, a friction angle, a few tuned constants	Costs almost nothing, runs on any elevation model, transparent	Constants for slip and resistance are not transferable (K varies about 75-fold, section 3.1); no sinkage; blind to sand state	Coarse screening large area ranking assumptic planning where entrapme the worst case
Classical Bekker-Wong, Wong-Reece	Bevameter or plate tests, shear tests, wheel geometry and load	Interpretable, closed-form, fast (SIM-028)	24 to 52 % median error on benchmark wheels, worse on firm soil (SIM-025 to SIM-027); parameters depend on plate geometry (TM-029); quasi-static (TM-018, TM-019)	Single-wh design tra studies or character soil, with tuning on soil
Granular RFT	Depth and orientation force law from a plate or intruder test	Few parameters; wheel drawbar mean absolute error 5.6 to 15 N and sinkage error 18 mm at 190 N on sand and fines (TM-001 to TM-004); relative errors of about 2.6 to 15 %, up to 30 % for reversed legs, in legged and plate tests (TM-008 to TM-013)	Error grows with load; response surfaces scaled from poppy seeds; no sloped-wheel field validation found	Slow quas static locomotio on dry gra in a controlled bed, legge robots, fas what-if studies

Family	Needs	Strengths (published)	Weaknesses (published)	Appropria when
Scaling laws	Matched wheel pairs	Collapse level-ground data within about 3 % (TM-007)	Slopes and low gravity unvalidated physically (TM-020 is DEM only)	Transferri a lab resu between wheel size on level sa
DEM	Particle contact calibration (friction, rolling resistance, cohesion), often scaled particles	Captures flow, avalanching and high-slip behaviour; about 13 to 18 % where a scalar error exists (SIM-018, SIM-021), with larger gaps for gross traction (20 against 31 % of weight, SIM-019)	Wall-clock hundreds to thousands of times simulated time with scaled particles on GPU (SIM-016, SIM-024) and about 9 million for fine particles on CPU (SIM-020), all derived; calibration is a tuning step, not independent evidence	Mechanis studies, generatin training da checking simpler models
Chrono SCM / CRM, multibody	Bekker-Wong constants (SCM) or density and friction angle (CRM)	Two to several orders of magnitude cheaper than DEM (SIM-007, SIM-008, SIM-011); BSD-3	Validation mostly plots, no scalar errors (SIM-001, SIM-002, SIM-009); fitted SCM constants can be unphysical (SIM-005)	Rover-sca simulation controller testing, generatin labelled d
Learning from images or terrain class	Slip labels per terrain from visual odometry or encoders	Held-out AUROC 0.874 on Perseverance (LRN-010)	Needs labelled runs on each terrain; class confusion inflates error (LRN-004); fails outside training roll/pitch (LRN-003)	Onboard hazard ranking where the vehicle ha driven sim sand
Learning from proprioception	Wheel speeds, currents, IMU	Detects slip as it happens (LRN-009, LRN-015, LRN-024)	Moderate slip is weakest (LRN-009); mostly binary events; random hold-outs	Online slip detection adaptation

Family	Needs	Strengths (published)	Weaknesses (published)	Appropria when
Hybrid or physics-informed learning	Physical model plus onboard adaptation	Online soil adaptation lifted physical grass-to-sand completion in a small trial (NeSAM, notes; the physical dataset size was not located)	Few physical tests, small trial counts	Promising adapting a planner to unseen sa
Empirical field tables (Mars slip tables)	Prior drives and operational constraints	Used in practice on Mars (FLD-020, FLD-021, FLD-024)	Failed on mixed terrain and hidden weak deposits (field notes)	Conserva operations with a hur in the loop

Selection rule used in this project. Use a cheap slope-and-slip rule to map where the answer is sensitive to assumptions (section 8), use a calibrated single-wheel model or RFT to choose wheel geometry on a characterised sand, and use proprioceptive or image-based learning on board only after the vehicle has logged enough runs on that sand. Use DEM only to understand mechanisms.

---

5. Open data for an Arabian dune planner

All retrievals 2026-10-11. Licences are as read from provider pages; legal reading is the user's responsibility. Full table: `research/raw/data_audit.csv`; licences: `docs/DATA_AND_LICENSES.md`.

Dataset	Resolution	Coverage	Licence (as read)
Copernicus DEM GLO-30 (DSM, TanDEM-X, 2010 to 2015)	1 arcsec; about 28.3 m E-W by 30.8 to 30.9 m N-S at 23.5 N (my arithmetic from tile geometry; the code uses 30.76 m)	Global land	Free with attribution (c) DLR and Airbus, EU/ESA; controlling licenc (COP-DEM-GLO-30-F) read in f phase 2: no commercial restrictio attribution and no-liability notices required (`docs/DATA_AND_LICENSES.`

Dataset	Resolution	Coverage	Licence (as read)
Copernicus GLO-90	3 arcsec (about 90 m)	Global land	As GLO-30
SRTM-derived "skadi" tiles	1 arcsec	Global (mixed SRTM and other sources per the tiles page)	Joerd project attribution, SRTM courtesy USGS
NASADEM, ASTER GDEM v3	1 arcsec	NASADEM 60 N to 56 S; ASTER 83 N to 83 S	Openly shared; download needs a Earthdata login
ALOS AW3D30 v4.1	1 arcsec	Global land within about 82 degrees latitude	Free after registration, JAXA term not read
FABDEM v1-2	1 arcsec	Global	CC BY-NC-SA 4.0 (non-comme
TanDEM-X 90 m (DLR)	3 arcsec	Global	Scientific use, registration
ICESat-2 ATL08, GEDI L2A	100 m segments; 25 m footprints	Tracks only; GEDI 54 N to 54 S	Free, Earthdata login
Sentinel-2 L2A	10, 20, 60 m	Global, frequent	Free, full and open
Landsat 8/9	30 m	Global	No restrictions stated

Dataset	Resolution	Coverage	Licence (as read)
Maxar Open Data	sub-metre class (figure not stated on the page read)	Event-based; no Gulf event found	CC BY-NC 4.0
Planet NICFI, Pleiades Neo	n/a	n/a	Planet terms not verified; Pleiade Neo commercial
Abu Dhabi SDI open data (habitat and geology polygons)	vector	Abu Dhabi emirate	Licence field empty in service metadata

Not found. Open airborne LiDAR or drone elevation models of Liwa or the Rub' al Khali dunes; any gridded sand colour or grain-size map of the Empty Quarter; any open sub-metre stereo for the region.

Can slip faces and crests be resolved? Evidence for the answer, all of it my arithmetic or computation unless marked published:

- Published dune sizes: Liwa dunes 50 to 80 m high (Lorenz et al., abstract); northern Rub' al Khali complex linear dunes 50 to 100 m (Lancaster et al., abstract); published mean wavelengths 0.63 to 3.17 km (USGS Professional Paper 1052 Landsat morphometry; `morphology_rows.csv`). At 30 m posting that is 20 to 110 posts per wavelength: well resolved. Small simple linear dunes of 10 to 25 m height and 200 to 300 m spacing (a secondary summary, to be re-verified) are 7 to 10 posts at 30 m and 2 to 3 at 90 m.
- A slip face at 32 degrees has a horizontal run of about 1.6 times its height, so a 50 m dune face spans about 80 m, 2.7 posts at 30 m. In the Rub' al Khali window, runs of consecutive posts steeper than 25 degrees had a median length of 2 posts (90th percentile 4; 45 % a single post).
- A synthetic test of an ideal 32 degree face sampled at 30 m with random phase (SIMULATED-BY-ME, optimistic because it samples points rather than averaging over cells) recovers full steepness only if the face is 30 m or higher by forward difference (50 m or higher by central difference). A 10 m face reads as 9 to 18 degrees and a 20 m face as 18 to 26 degrees. At 90 m, even 50 m faces read 16 to 23 degrees.
- **Verdict.** Dune crests, ridges and interdune corridors of the large Liwa and Rub' al Khali dunes: resolved at 30 m. Slip faces of dunes above about 40 to 50 m: marginal, located as thin bands of high slope but biased low. Slip faces of dunes below about 20 m, brink sharpness, ripples (centimetre wavelength) and small barchans of 1 to 7 m: not resolved, and the barchans are at or below the published vertical error. A planner reading 30 m slope alone will under-flag lee faces.
- **Caveats.** GLO-30 is a radar surface model; the size of X-band bias over dry sand was not retrieved. The epoch (2010 to 2015) is stale for fast small dunes (over 20 m/yr in Kuwait). No lidar truth was available, so

slip-face bias is estimated by synthetic and cross-product comparison only.

---

6. UAE context (factual background; every item checked against retrieved sources, see `research/raw/uae_facts.csv`)

- **Rashid 1** (MBRSC, about 10 kg, four wheels, designed for slopes up to 20 degrees and rocks 10 cm high) never operated: the ispace Hakuto-R Mission 1 lander failed to land on 2023-04-26 (JST), which ispace attributed to altitude-sensor handling in software (UAE01, UAE02, UAE10, UAE11).
- **Rashid 2** is built and was shipped to the US for Firefly's Blue Ghost Mission 2 (lunar far side). Launch dates conflict between sources (2026 in late-2025 coverage, not earlier than 2027 in an October 2026 report), so no flight data exist and the date is unresolved (UAE14, UAE15, UAE17).
- **Wheel work.** Khalifa University and MBRSC published single-wheel rig results for the Rashid wheel on a lunar regolith sample (slip 0 to 0.75, sinkage about 1 to 2 cm, drawbar 5.2 to 24.1 N, UAE04, UAE06) and wheel and grouser studies (UAE07, UAE08). These concern lunar simulant, not Arabian sand.
- **Desert ground vehicles.** A TII desert unmanned ground vehicle (page date ambiguous, 2023 or 2025) and, at UMEX in January 2026, SteerAI's xRift (debuted 20 January 2026) and a Space42 vehicle (25 degree slope capability stated) were shown; the retrieved pages give no slip, sinkage or dune-traversability data (UAE22 to UAE25).
- **Programmes that are not mobility evidence.** Hope is an orbiter (UAE19, UAE20); Mars Science City was announced as a simulated settlement with 3D-printed desert-sand domes and its article does not mention rover testing (UAE21); the Emirates Robotics Competition task is indoor (UAE26).
- **Not verified or not found:** Rashid 2 wheel design, current Rashid 2 launch date, any rover testing on UAE dunes by UAE universities, current Mars Science City status. The 1963 archive item on wheeled-vehicle sand trials could not be read.
- Thermal and dust effects on these vehicles are out of scope; see `hudasol/dhill`.

---

7. Gap analysis

7.1 Where lab-to-field transfer fails or is untested

1. **Every quantitative sand-slope result is lab, a prepared Earth sand bed (SIM-012, SIM-013; the unit of the slope term is not stated in its source) or Mars; none is Arabian dune sand.** Lab slope beds use simulants or Toyoura sand, mostly small wheels (exception: the 400 kg Scarab, FLD-027, FLD-028) and slow speeds (FLD-027 to FLD-035, LEG-017 to LEG-020, TM-022 to TM-024). Mars field data are low-gravity and on megaripples, crusts and sulfates (FLD-001 to FLD-025, apart from the Earth tilt-bed rows FLD-006 to FLD-008).
2. **Engineering tables transferred from a lab bed work on uniform sand and fail on mixed or hidden weak ground** (field notes, Maimone et al.). The Spirit pair (15 to 20 % versus 95 % slip on terrain with no visible difference, FLD-011, FLD-012) shows the pattern; the source does not state the cause.
3. **Model validation uses in-sample or borrowed parameters.** The rover-scale slope error fell from 25.9 to 0.49 % only after tuning on the same case (TM-023, TM-024); DEM and SCM fits are calibrations (SIM-005, SIM-015, SIM-018).
4. **The quasi-static models are outside their validated envelope when driving is fast or transient** (TM-018, TM-019, TM-021).
5. **Learning evidence is in-distribution.** Hold-outs are random, temporal or by sol, never by terrain type (LRN-003, LRN-007, LRN-009, LRN-015). The one held-out-soil result is a simulation (LRN-016).
6. **The field slip sensor is unreliable on exactly the terrain of interest.** Visual odometry degrades on featureless sand (FLD-041), and different papers define slip differently (wheel odometry, visual odometry, velocity-based, distance-based), so numbers are not interchangeable.

7.2 What is missing for dune terrain specifically

1. A **slip-versus-slope dataset with a wheeled vehicle on dune sand with complete soil and wheel description** and independent ground-truth position. None found.
2. **Behaviour on and near the slip face**: avalanching and shear collapse near the angle of repose. Slope-induced shear collapse is shown only for a foot on silica sand (sinkage 13 mm at 0 degrees to 36 mm at 35 degrees, SIM-022, SIM-023) and in a few legged tests. Simulators that omit it will under-predict.
3. **Arabian sand properties**: grain size, rounding, packing, moisture and cementation below the surface, and Bekker or RFT constants. Nearest values are Kuwaiti barchans (mean grain 0.23 mm) and a Brazilian dune (d50 about 0.2 mm), not Gulf dunes (`morphology_rows.csv`).

4. **Dune-scale terrain data at the resolution that matters** (open, metre-scale elevation). None open for the region; GLO-30 under-resolves slip faces (section 5).
5. **A common slip definition and a shared benchmark** so that rows can be compared.
6. **Dynamic effects**: crosswind and lateral slip on crests, slope at speed, repeated passes on the same track (not covered by any retrieved row).
7. **Tracked and legged vehicles on real dunes**, and any same-vehicle comparison of the three.

7.3 What a minimal open-data planner needs that the literature does not provide

- A slip law whose constants come from **measurements on the target sand and the target vehicle**, with stated uncertainty. This project supplies the structure and an analysis plan (sections 8 and 9) but not the constants.
- A **slip-face and crest detector** that does not rely on a 30 m slope value: aspect, curvature, dune height and optical imagery are the open candidates, but none has been validated here.
- A **way to carry the assumption uncertainty into the plan**, for example by treating a route as acceptable only if it is acceptable across the plausible parameter range, which is what the sensitivity analysis in section 8 suggests is necessary.
- An **onboard log** (section 9.5) that lets the vehicle correct the model as it drives.

---

8. Simulation study (SIMULATED-BY-ME): slope-and-slip traversability from an open elevation model

> **SIMULATION ONLY, model-assumption dependent. Nothing in this section is field evidence.** The model is a deliberately minimal planning screen, not a vehicle model. It omits sinkage, wheel geometry, vehicle mass, speed, avalanching and cohesion. The published evidence in section 3 shows that the slip it tries to represent varies from near 0 to near 1 at similar slopes, so the outputs below describe the **model**, not any real rover on any real dune.

8.1 What was run

- **Data.** Copernicus DEM GLO-30 tiles N23E053 and N22E054 (licence in section 5; tile hashes in `results/summary.json`). Three windows: **A-south Liwa** (23.02 to 23.17 N, 53.55 to 53.95 E, about 17 by 41 km, relief 139 m),

A-north Liwa (23.25 to 23.45 N, same longitudes; smaller crescentic dunes, relief 85 m) and **B Rub' al Khali megadunes** (22.20 to 22.50 N, 54.20 to 54.60 E, relief 211 m). The windows were chosen by me for visible dune texture, not at random.

- **Slope.** Horn 3 by 3 gradient with per-latitude metric spacing (`src/dunerover/terrain.py`). A central-difference check on the same A-south window gave 1.70 % of cells above 20 degrees in the earlier data audit against 1.45 % here (Horn): slope tails depend on the kernel by roughly 15 % relative.
- **Slip model** (`src/dunerover/slipmodel.py`). For an edge between neighbouring cells with along-heading slope angle theta_a and cross slope theta_c: demand D = sqrt((tan theta_a + f_r)^2 + tan^2 theta_c) (friction circle), capacity mu_max = tan(phi), slip i = -i0 ln(1 - D / mu_max), entrapment if D >= mu_max. An edge is passable if i <= i_max; travel cost = distance / (1 - i). The exponential comes from the Janosi-Hanamoto shear law applied at the contact-patch level.
- **Planner.** 8-neighbour directed graph, Dijkstra least-cost routes (`scipy.sparse.csgraph`), ten deterministic origin-destination pairs per window picked on gentle low ground (seed 20261011). Slope in each direction is taken from the elevation difference along the edge, cross slope from the Horn gradient at the edge midpoint.
- **Seeds and reproducibility.** Origin-destination 20261011; Latin hypercube 20261012 (A core) and 20261013 (B core); 256 draws each, on core windows of 0.07 by 0.16 degrees (A-core 23.06 to 23.13 N, 53.62 to 53.78 E; B-core 22.28 to 22.35 N, 54.30 to 54.46 E); the full study was run twice and the area and sensitivity entries of the two `results/summary.json` files were identical (a one-off comparison that is not stored in the repository; re-run to confirm). Runtime about 330 s.

Table 8.1: parameters of the slip model. Only the friction angle has a published basis; the other three are assumptions with ranges. DEM noise is an additional assumption used in the sensitivity analysis only (spatially correlated, correlation length 3 posts, standard deviation 0 to 2 m).

Parameter	Nominal	Range	Status	Basis
`phi_deg`	33.0	28.0 to 38.0	published range of laboratory sands (not measured for Arabian sand)	Internal friction angles printed in retrieved papers: 28 (Irani dry sand), 34 (Quikrete sand), 35 (Mojave simulant), about 35 (olivine sand), 38 (Toyoura, Tottori); angle of repose 32-35 deg (Kroy et al. 2002). See research/raw/terramechanics_notes.md and morphology_rows.csv (M05).
`f_r`	0.12	0.05 to 0.25	ASSUMPTION	Motion-resistance coefficient (resistance/weight) lumped for a small wheeled UGV in loose sand. No retrieved source gives a transferable value; range chosen to span light-to-heavy sinkage while keeping the vehicle able to move on flat sand. Replace with a measured value for the real vehicle.
`i0`	0.2	0.05 to 0.4	ASSUMPTION	Slip scale K/L. Published Janosi shear modulus K spans 0.0006-0.045 m between papers (terramechanics_notes.md), contact length L is vehicle specific, so i0 is a tuning parameter.

Parameter	Nominal	Range	Status	Basis
`i_max`	0.5	0.4 to 0.7	ASSUMPTION (planning threshold)	Slip above which an edge is treated as impassable. A Mars rover autonav slip table needed limits above 50 % on 15-20 deg terrain (evidence FLD-024); that is Mars-specific, used only as a pointer.
`noise_sigma_m` (sensitivity only)	0	0 to 2 m	ASSUMPTION	Spatially correlated DEM error, correlation length 3 posts; no measured value for dune DEM error was retrieved.

8.2 Elevation statistics (computed from the data, not from the slip model)

Window	Posting, m	Block	Relief, m	Max slope, deg	% cells >10	>15	>20	>25	>30
A_south_Liwa	29.6	1	139	37.0	14.6	3.89	1.45	0.57	0.0549
A_south_Liwa	59.3	2	136	29.5	6.86	2.04	0.632	0.0931	0
A_south_Liwa	88.9	3	133	25.5	5.23	1.25	0.163	0.00463	0
A_north_Liwa	29.6	1	85	35.1	15.3	4.34	1.39	0.304	0.0242
A_north_Liwa	59.2	2	81	26.1	6.52	0.917	0.061	0.000386	0
A_north_Liwa	88.8	3	77	19.4	1.72	0.0365	0	0	0
B_RubAlKhali_megadunes	29.7	1	211	42.7	33.1	12.6	4.97	2.33	0.45
B_RubAlKhali_megadunes	59.4	2	205	33.8	25.4	7.94	3.09	1.15	0.0283
B_RubAlKhali_megadunes	89.1	3	200	29.9	20.8	5.86	1.91	0.437	0

Coarsening to 60 and 90 m (block mean) removes most steep cells: the share of A-south cells above 25 degrees falls from 0.57 % at about 30 m to 0.093 % at about 60 m and 0.0046 % at about 90 m. This is the resolution effect discussed in section 5, reproduced on real tiles.

8.3 Nominal results

Under nominal parameters the model's critical direct-ascent slope (slip equal to i_max) is **25.5 degrees**; the slip curve reaches entrapment at about 28 degrees. Model slip at 0, 5, 10, 15, 20 and 25 degrees of direct ascent is about 0.04, 0.08, 0.12, 0.18, 0.27 and 0.47.

Window	Grid (rows x cols)	Blocked edges	Reachable share	Median cost ratio	Median length ratio	Steepest uphill on any route, deg	Origin-destination pairs
A_south_Liwa	541 x 1441	0.25 %	1.0000	1.116	1.063	20.4	10 (0 unreachable)
A_north_Liwa	721 x 1441	0.20 %	1.0000	1.115	1.064	18.1	10 (0 unreachable)
B_RubAlKhali_megadunes	1081 x 1441	0.97 %	0.9999	1.115	1.056	23.1	10 (0 unreachable)

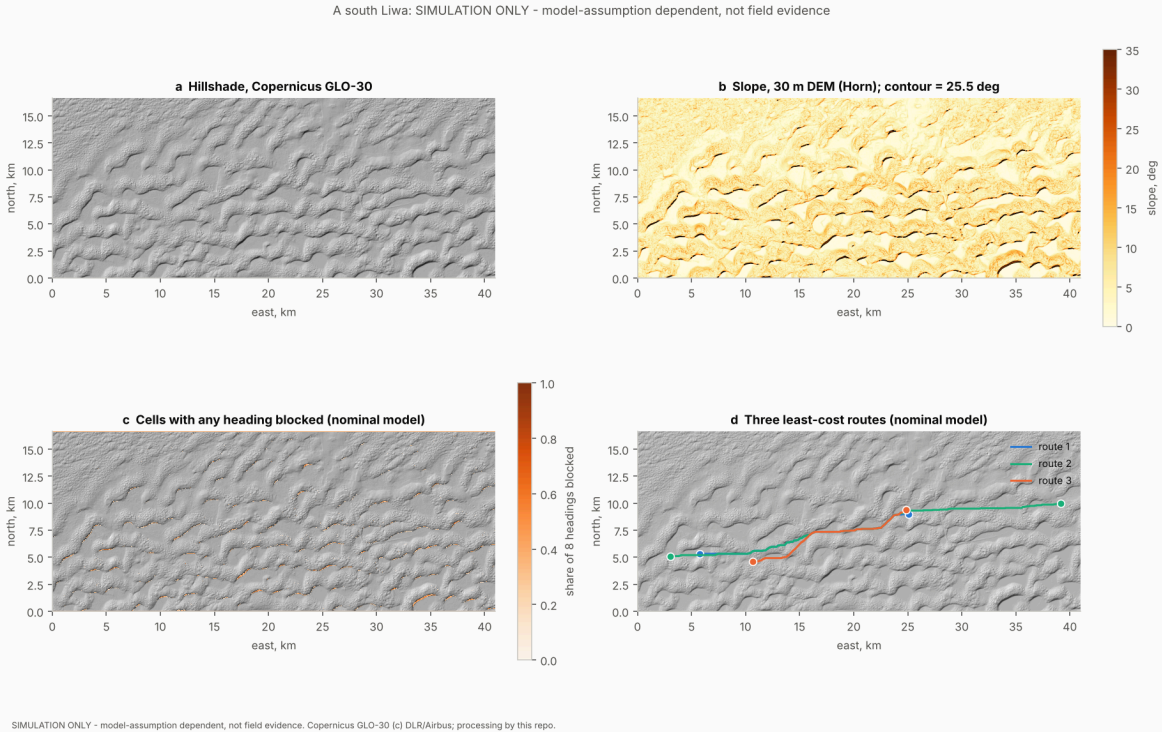


Figure 8.1. (a) hillshade of GLO-30, (b) slope with the nominal critical angle contoured, (c) cells where at least one of eight headings is blocked, (d) three least-cost routes. SIMULATION ONLY, model-assumption dependent, not field evidence. The Rub' al Khali window is `results/figures/fig_area_B_RubAlKhali_megadunes.png`.

How to read these numbers.

- The blocked share is tiny (0.2 to 1 % of edges). That follows from two things, not from the dunes being easy: at 30 m posting only a small share of cells exceed the nominal critical angle (section 8.2), and slip-face slopes are biased low by the grid (section 5). A real rover facing the steepest faces would be blocked far more often than this map shows.
- The median route cost ratio (about 1.115 in all three windows) is **almost entirely a grid and baseline effect**: an 8-neighbour path is about 6 % longer than the straight line (column "Median length ratio" above) and nominal flat-sand slip of about 0.04 adds a factor near 1.04. It should not be read as a dune effect. The routes avoid steep cells at small cost because the origin and destination are on gentle ground.
- Reachability is about 0.9999 in every window: the interdune corridors connect everything at nominal parameters.

8.4 Sensitivity analysis: which assumptions change the map most

Two analyses were run. **One-at-a-time (OAT):** each parameter to its low and high value, plus structural alternatives (slip-law form, cross-slope rule, DEM posting, DEM noise). **Global:** 256 Latin-hypercube draws of the five continuous assumptions on a core window of A and of B, with Spearman

rank correlations against each output (ranks are used because the outputs are monotonic but non-linear; the rank regression R-squared values in `results/summary.json` are 0.86 to 0.96 for blocked share, reachability and climbing limit, so the ranking explains most of the variance).

Table 8.2: one-at-a-time sensitivity, area A-south (SIMULATED-BY-ME).

Case	A-south blocked share	x nominal	A-south median cost ratio	B blocked share	B median cost ratio	Critical uphill, deg
nominal	0.0025	1	1.116	0.0097	1.115	25.5
phi_deg=28.0	0.0084	3.38x	1.132	0.0271	1.139	20.2
phi_deg=38.0	0.0002	0.07x	1.106	0.0009	1.103	30.8
f_r=0.05	0.0010	0.40x	1.093	0.0051	1.093	28.6
f_r=0.25	0.0083	3.35x	1.186	0.0243	1.188	19.1
i0=0.05	0.0011	0.44x	1.076	0.0048	1.070	27.9
i0=0.4	0.0110	4.43x	1.177	0.0344	1.198	18.9
i_max=0.4	0.0038	1.51x	1.116	0.0136	1.115	23.8
i_max=0.7	0.0015	0.61x	1.116	0.0064	1.115	27.0
form=janosi_mean	0.0199	8.02x	1.185	0.0561	1.217	16.2
cross=additive	0.0097	3.92x	1.126	0.0325	1.134	25.5
dem_block=2	0.0006	0.25x	1.114	0.0049	1.113	25.5
dem_block=3	0.0001	0.05x	1.113	0.0023	1.111	25.5
dem_noise=1.0 m	0.0025	1.00x	1.116	0.0096	1.115	25.5
dem_noise=2.0 m	0.0025	1.01x	1.117	0.0096	1.115	25.5

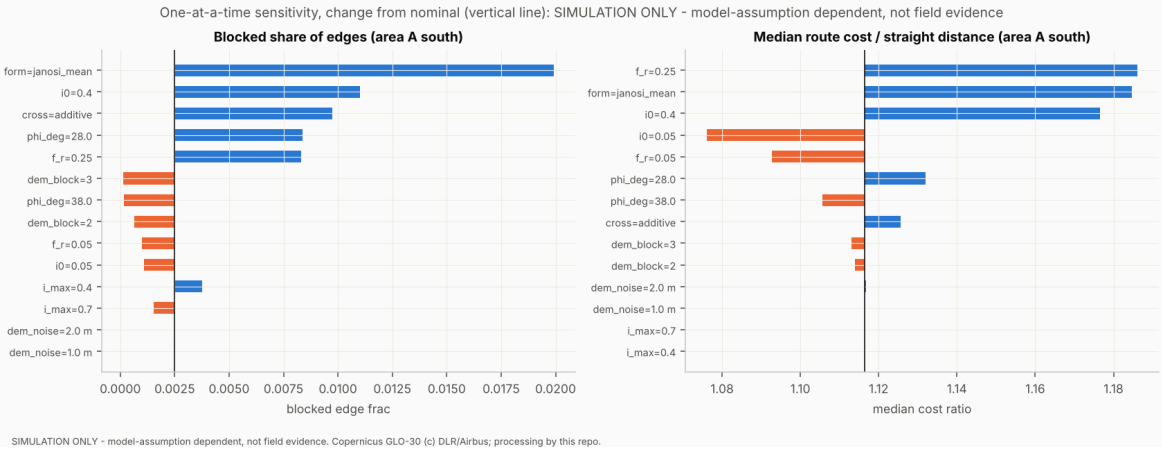


Figure 8.2. Change from nominal in blocked share (left) and median route cost ratio (right), area A-south. SIMULATION ONLY.

Table 8.3: global sensitivity, Spearman rank correlation of each assumption with each output (Latin hypercube, n = 256, SIMULATED-BY-ME).

Assumption	A: blocked share	A: cost ratio	A: critical uphill	B: blocked share	B: cost ratio	B: critical uphill
friction angle phi	-0.67	-0.29	+0.63	-0.69	-0.26	+0.63
resistance f_r	+0.46	+0.58	-0.53	+0.36	+0.53	-0.51
slip scale i0	+0.56	+0.75	-0.54	+0.59	+0.79	-0.54
slip limit i_max	-0.27	-0.04	+0.26	-0.12	+0.08	+0.08

Assumption	A: blocked share	A: cost ratio	A: critical uphill	B: blocked share	B: cost ratio	B: critical uphill
DEM noise	+0.06	+0.02	-0.05	-0.11	-0.08	+0.10

Table 8.4: spread of outputs across the assumption ranges (5th, 50th and 95th percentile, and extremes; SIMULATED-BY-ME).

Core window	Output	min	5 %	median	95 %	max
A_core	blocked share	0.0000	0.0002	0.0043	0.0240	0.0509
A_core	median cost ratio	1.079	1.087	1.133	1.272	1.369
A_core	reachable share	0.9964	0.9994	1.0000	1.0000	1.0000
A_core	critical uphill, deg	11.3	14.7	23.4	31.4	34.7
B_core	blocked share	0.0001	0.0003	0.0132	0.0550	0.1950
B_core	median cost ratio	1.090	1.099	1.152	1.295	1.581
B_core	reachable share	0.9757	0.9895	1.0000	1.0000	1.0000
B_core	critical uphill, deg	7.6	15.3	23.4	31.6	34.5

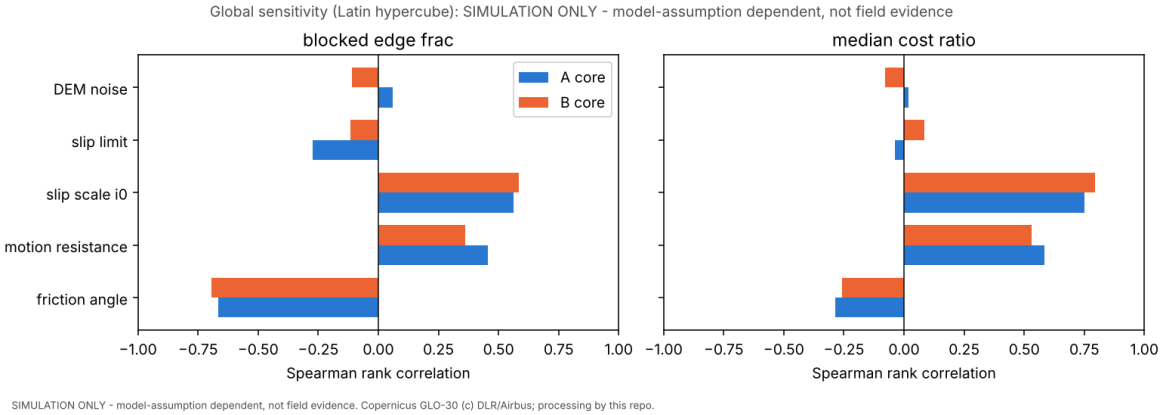


Figure 8.3. Spearman rank correlations (Latin hypercube). SIMULATION ONLY.

Reading of the sensitivity results.

1. **Structural modelling choices move the map as much as the parameter ranges.** Replacing the point-wise exponential with the contact-patch-averaged Janosi form raises the blocked share in A-south from 0.0025 to 0.0199 (8-fold, the largest single upward change in Table 8.2) and lowers the model's critical slope to 16.2 degrees (independent of the DEM). The additive cross-slope rule (instead of the friction circle) raises the blocked share about 4-fold. Neither choice has published validation for dune sand. For comparison, moving the friction angle from 28 to 38 degrees changes the blocked share about 50-fold (0.0084 to 0.0002), so the friction angle gives the largest swing of any single parameter.
2. **DEM posting matters as much as the parameters.** Block-averaging to 60 m and 90 m cuts the blocked share to 0.25 and 0.05 times nominal in A-south and to about 0.5 and 0.24 times in B (Table 8.2), because slip faces disappear (section 5). A map built from a coarser product would look easier than the 30 m map, and the 30 m map is itself likely too easy.

3. **Among the continuous parameters, the slip scale i0, the friction angle and the resistance coefficient dominate.** In both cores, the friction angle has the strongest rank correlation with blocked share (about -0.67 to -0.69), then i0 (+0.56 to +0.59), then f_r (+0.36 to +0.46). For the route cost ratio, i0 and f_r come first. The model's climbing limit spans 7.6 to 34.7 degrees across the draws (5 to 95 %: 14.7 to 31.4 in A, 15.3 to 31.6 in B). Two of these three (i0, f_r) are assumptions with no transferable published value, and the third (phi) differs by 10 degrees between published lab sands, so this is where measurements pay off most.
4. **The slip limit i_max and spatially correlated DEM noise matter least** in this setup (rank correlation magnitudes at most 0.27 and 0.11). The noise result is conditional on the assumed noise model; real DEM error over dunes (X-band penetration, slip-face bias) is not modelled.
5. **Blocked share varies by about two orders of magnitude** between the 5th and 95th percentiles of the assumption space (A core 0.02 to 2.4 %; B core 0.03 to 5.5 %), with extremes of 0 to 5 % (A) and 0.01 to 19.5 % (B), while the OAT ranking is similar in the two areas. The conclusion that matters for a planner is that **the map is controlled by assumptions that field data have not yet measured**.

8.5 Comparison with published slip (descriptive only, nothing fitted)

Thirteen published slip-versus-slope points were transcribed by hand (`data/anchor_points.csv`: nine from FLD-010, FLD-011, FLD-012, FLD-025 and FLD-027 to FLD-031, and four from LEG-017 to LEG-020). They differ in slip definition, gravity, wheel and soil (see the warning column) and **should not be used as a test of the model**. The comparison shows only whether the assumption envelope overlaps them.

Row	Slope, deg	Published validation	Published slip	Model nominal (SIM)	Model envelope min to max (SIM)	Envelope overlaps	Definition warning
FLD-010	11.5	field	0.00 to 0.50 (upper_bound)	0.14	0.02 to 0.67	yes	slip = 1 - achieved/commanded distance; Mars; slope range; 'up to 50 %' used as an upper bound
FLD-011	19.0	field	0.15 to 0.20 (range)	0.25	0.04 to 1.00	yes	VO-based slip; Mars; appearance-identical terrain to FLD-012
FLD-012	16.0	field	0.95	0.20	0.03 to 1.00	yes	VO-based slip; Mars
FLD-025	20.0	field	0.94 to 1.00 (lower_bound)	0.27	0.04 to 1.00	yes	rover-level VO slip; slope 'about 20'; '>94 %' used as a lower bound
FLD-027	20.0	lab	0.90 to 1.00 (lower_bound)	0.27	0.04 to 1.00	yes	velocity-based slip; lunar simulant, rigid wheels without grousers; '>90 %' used as a lower bound

Row	Slope, deg	Published validation	Published slip	Model nominal (SIM)	Model envelope min to max (SIM)	Envelope overlaps	Definition warning
FLD-028	25.0	lab	1.00	0.47	0.06 to 1.00	yes	no uphill progress at 25 deg direct ascent, treated as slip = 1
FLD-029	15.0	lab	0.20 to 0.30 (range)	0.18	0.03 to 1.00	yes	descriptive capability statement; lunar simulant
FLD-030	4.0	lab	0.85 to 0.95 (range)	0.07	0.01 to 0.33	NO	text says close to 0.9; band +/-0.05 is mine
FLD-031	4.0	lab	0.15 to 0.25 (range)	0.07	0.01 to 0.33	yes	text says around 0.2; band +/-0.05 is mine
LEG-017	12.0	lab	0.74	0.14	0.02 to 0.71	NO	slope emulated by horizontal pull; slides not a paper
LEG-018	12.0	lab	0.06	0.14	0.02 to 0.71	yes	slope emulated by horizontal pull; diameter and lug height varied together
LEG-019	14.0	lab	0.47	0.17	0.03 to 0.93	yes	slope emulated by horizontal pull
LEG-020	14.0	lab	0.25	0.17	0.03 to 0.93	yes	slope emulated by horizontal pull

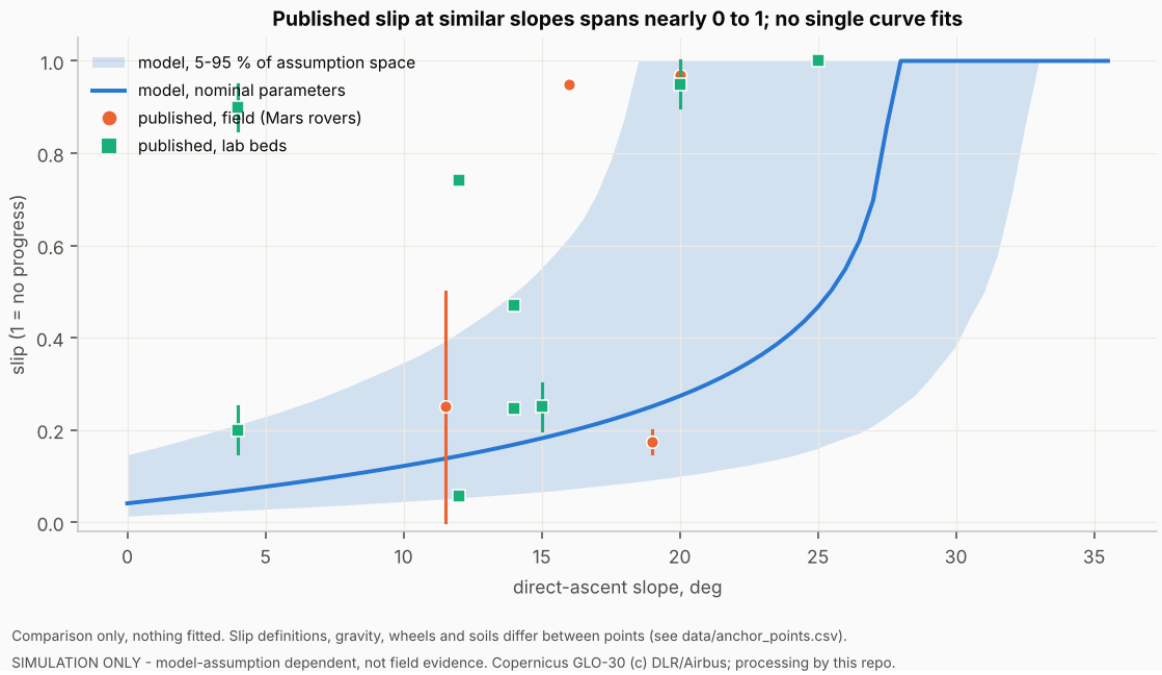


Figure 8.4. Blue line and band: model at nominal parameters and its 5 to 95 % range over the assumption space (SIMULATED-BY-ME). Orange circles: published field results (Mars rovers). Green squares: published lab beds. Bars show reported ranges or bounds. Published points at similar slopes span nearly 0 to 1 and no single curve fits them.

The assumption envelope overlaps 11 of the 13 published points, which says mostly that it is wide, not that the model is valid. The nominal curve is more than 0.5 slip away from six of them (FLD-012, FLD-025, FLD-027, FLD-028, FLD-030, LEG-017), and the envelope misses two (FLD-030, a smooth-

wheeled lab rover at 0.9 slip at 4 degrees, and LEG-017, a 116 mm wheel at 0.741 at 12 degrees). These are cases a slope-only model with lumped constants cannot represent.

8.6 Limitations specific to this study

- Quasi-static, point-mass, no sinkage, speed or avalanching; no wheel geometry or vehicle mass; i0 and f_r absorb everything.
- Slope is taken from a 30 m radar surface model whose slip-face slopes are biased low (section 5); the blocked shares of phases 1 are lower bounds in that sense. Section 8.7 adds a modelled sub-grid face probability and an UNKNOWN class to address this, with its own assumptions.
- The windows are not random samples of any dune field. Origins and destinations sit on gentle low ground, so route statistics say nothing about crossing slip faces.
- Descent is treated with the same symmetric friction-circle demand ($|\tan\theta_a + f_r|$); descending stability and avalanching are not modelled.
- Parameters were not fitted to any data. No claim is made that the nominal values represent Arabian sand or any vehicle.
- Code is tested (`tests/test_core.py`: 14 tests covering exact planar slope, aspect, slip monotonicity and entrapment, cross-slope behaviour, graph blocking, flat-ground cost ratio, determinism of origin-destination pairs and sampling, coarsening, noise reproducibility and metres-per-degree), but passing tests show the code does what it says, not that the model is right.

8.7 Phase 2: what the elevation model cannot see (SIMULATED-BY-ME, model-assumption dependent)

> **SIMULATION ONLY, model-assumption dependent.** Everything below describes a synthetic detector and a model, validated only against synthetic truth. No lidar, drone or field profile of a real Arabian slip face was retrieved, so the detector has never been tested against reality. The predictions for this stage were committed before the code was run (`docs/PREDICTIONS.md`, stage 2) and the hits and misses are recorded there.

Problem. A 30 m GLO-30 post spacing cannot resolve an active slip face, whose horizontal run is only about 1.4 to 1.7 times its height (morphology row PM10): a few metres to a few tens of metres. A cell that contains a face reads as a gentle slope. The phase 1 map was therefore binary and optimistic.

Sub-grid face probability. For each cell I estimate the probability that an unresolved active face (28 degrees or steeper) lies inside the cell from four features the DEM does see: slope, crest convexity along the steepest direction, slope break (largest change of the signed slope component between the cell and its eight neighbours) and local relief in a 7 by 7 window (a proxy for dune height). The mapping is a logistic regression fitted on a synthetic ensemble of straight crested asymmetric dunes (stoss, brink, active face, apron, flat interdune) that are sampled the way a DEM samples them (mean over a post, or a point sample), separately for each posting (`src/dunerover/subgrid.py`). Truth in the ensemble is known exactly. Every range used is in Table 8.7a; only the stoss slope, the active face slope and part of the face height fraction have a published basis (retrieved morphology rows), and the rest are labelled assumptions. The share of dune crests that carry an active face is **not published** (not found), so the baseline assumes all of them do (pi = 1, the worst case) and the result is shown for pi = 0.6 and 0.3.

Table 8.7a: ranges of the sub-grid ensemble.

Sub-grid quantity	Range used	Status	Basis (morphology row ids in research/phase2/phase2_morphology.csv)
`stoss_slope_deg`	6.0 to 20.0	published (barchan and transverse dunes, small and medium)	PM01 (barchan 8 to 20), PM02/PM06 (transverse about 6). No distribution for dunes of tens of metres was found, so extrapolated.
`face_slope_deg`	29.5 to 32.5	published (lidar, two US fields)	PM18: sharp peak at 32.5 deg (White Sands) and 29.5 deg (Great Sand Dunes). Not shown to hold in Arabia.
`face_height_fraction`	0.2 to 1.0	partly published, mostly ASSUMPTION	PM08: 0.54 to 1.0 for six transverse dunes of 7 to 10 m. The 0.2 lower end is an assumption for tall compound dunes (no source found).
`dune_height_m`	5.0 to 120.0	published range, log-uniform is an ASSUMPTION	PM29 to PM35: Rub' al Khali dunes from under 5 m to over 155 m; 50 to 130 m for the linear and star types.
`interdune_flat_ratio`	0.0 to 2.0	ASSUMPTION	Flat interdune width as a multiple of the dune footprint; no source used.
`apron_slope`	equal to stoss slope	ASSUMPTION	Slope of the slip face base below the face; no source.
`no_face_lee_slope_deg`	10.0 to 25.0	ASSUMPTION	Lee slope when the face is inactive; the morphology search found no source for it (PM20).
`p_active_face (pi)`	0.3 to 1.0	ASSUMPTION	Share of dune crests with an active face; no source found. Baseline 1.0 (worst case).

Three classes. For each of 256 Latin hypercube draws of the four assumed slip parameters (friction angle, f_r, i0, i_max; fixed seeds 20261014, 20261015 and 20261016) a cell is passable if at least 4 of its 8 headings are passable; q is the share of draws in which it is passable. **BLOCKED** if q is at most 0.05. **TRAVERSABLE** if q is at least 0.95 and the face probability is below tau = 0.20. **UNKNOWN** otherwise, with the cause recorded: assumption sensitive (q between 0.05 and 0.95), sub-grid face risk (face probability at least tau), or both. The threshold tau is a free choice; the coverage versus certainty table shows what it trades. Third area **C** is a

window of linear dunes in the Saudi Arabian Rub' al Khali (20.40 to 20.55 N, 52.30 to 52.70 E, tile N20E052; not in the UAE), chosen because the UAE tiles held here show star and network dunes. It has a different morphology from windows A and B and is a robustness check, not a study of the UAE.

Table 8.7b: class shares by DEM posting. Coarser postings are block means of the 30 m grid (a stand in for GLO-90; the real GLO-90 product differs).

Area	Posting, m	TRAVERSABLE	BLOCKED	UNKNOWN total	UNKNOWN: assumption sensitive only	UNKNOWN: sub-grid face risk only	UNKNOWN both
A_south_Liwa	29.6	91.4 %	0.00 %	8.6 %	0.0 %	7.8 %	0.8 %
A_south_Liwa	59.2	90.9 %	0.00 %	9.1 %	0.0 %	8.9 %	0.2 %
A_south_Liwa	88.8	92.9 %	0.00 %	7.1 %	0.0 %	7.0 %	0.0 %
B_RubAlKhali_megadunes	29.7	89.1 %	0.01 %	10.9 %	0.0 %	7.9 %	2.9 %
B_RubAlKhali_megadunes	59.4	80.1 %	0.00 %	19.9 %	0.1 %	18.3 %	1.5 %
B_RubAlKhali_megadunes	89.1	85.6 %	0.00 %	14.4 %	0.0 %	13.6 %	0.7 %
C_RubAlKhali_linear	29.9	97.5 %	0.00 %	2.5 %	0.0 %	2.5 %	0.0 %
C_RubAlKhali_linear	59.7	89.8 %	0.00 %	10.2 %	0.0 %	10.2 %	0.0 %
C_RubAlKhali_linear	89.6	72.5 %	0.00 %	27.5 %	0.0 %	27.5 %	0.0 %

At 30 m the UNKNOWN share is 8.6 % in A south Liwa, 10.9 % in B (megadunes) and 2.5 % in C (linear dunes); BLOCKED is 0.00 %, 0.01 % and 0.00 %. The BLOCKED class is almost empty because the 30 m map hardly contains slopes the model calls impassable in every draw: that is a statement about what the DEM can show, not about the dunes (section 5 and the sub-grid problem above). At 60 and 90 m the UNKNOWN share is 9.1 % and 7.1 % (A), 19.9 % and 14.4 % (B), 10.2 % and 27.5 % (C). It is not monotone in A, because coarser cells average out relief and slope break and the detector, trained per posting, also shifts.

How much of the map is robust to the four assumed parameters. Slope based verdicts (before the sub-grid step) are unanimous across the draws for 99.1 % (A), 97.0 % (B) and 100.0 % (C) of cells at 30 m; only 0.85 %, 2.96 % and 0.00 % are sensitive to them. The slip parameters therefore matter far less for the three class map than the **resolution and the face question**, because almost every 30 m cell is gentle enough to pass under every assumed parameter set. This does not contradict the phase 1 sensitivity result (the climbing limit itself varies from 7.6 to 34.7 degrees across the ranges): the DEM simply rarely shows slopes near that limit.

Table 8.7c: robustness to the four assumed slip parameters.

Area	Posting, m	Robust: same verdict in at least 95 % of draws	Sensitive to the four assumptions	Passable in all 256 draws	Passable in no draw
A_south_Liwa	29.6	99.15 %	0.85 %	98.33 %	0.00 %

Area	Posting, m	Robust: same verdict in at least 95 % of draws	Sensitive to the four assumptions	Passable in all 256 draws	Passable in no draw
A_south_Liwa	59.2	99.82 %	0.18 %	99.50 %	0.00 %
A_south_Liwa	88.8	99.95 %	0.05 %	99.88 %	0.00 %
B_RubAlKhali_megadunes	29.7	97.04 %	2.96 %	94.95 %	0.00 %
B_RubAlKhali_megadunes	59.4	98.39 %	1.61 %	97.00 %	0.00 %
B_RubAlKhali_megadunes	89.1	99.25 %	0.75 %	98.23 %	0.00 %
C_RubAlKhali_linear	29.9	100.00 %	0.00 %	99.95 %	0.00 %
C_RubAlKhali_linear	59.7	100.00 %	0.00 %	99.99 %	0.00 %
C_RubAlKhali_linear	89.6	100.00 %	0.00 %	99.99 %	0.00 %

Sensitivity to the sub-grid assumptions. The UNKNOWN share (30 m) when the share of crests with an active face falls from 1.0 to 0.3 is 2.9 % (A), 5.9 % (B) and 0.2 % (C). The most important choice is the definition of "inside the cell". The pre-registered definition counts a face only if it lies within the cell itself. A **post hoc variant** (added after the baseline result was seen, so not pre-registered) counts a face within one post of the cell, which is closer to the question a route cares about; it gives UNKNOWN shares of 34.4 % (A), 32.2 % (B) and 22.0 % (C) at 30 m. Both are shown because the honest range is wide.

Table 8.7d: UNKNOWN share under the sub-grid variants.

Area (30 m)	Sub-grid variant	pi (share of crests with an active face)	UNKNOWN total	Mean face probability
A_south_Liwa	base	1.0	8.6 %	7.3 %
A_south_Liwa	base	0.6	5.0 %	4.9 %
A_south_Liwa	base	0.3	2.9 %	3.1 %
A_south_Liwa	gentle	1.0	12.3 %	8.9 %
A_south_Liwa	steep	1.0	8.6 %	7.8 %
A_south_Liwa	POST HOC: face within +/-1 post (3-post footprint)	1.0	34.4 %	21.7 %
A_south_Liwa	base, point sampling	1.0	8.2 %	7.1 %
B_RubAlKhali_megadunes	base	1.0	10.9 %	9.6 %
B_RubAlKhali_megadunes	base	0.6	8.1 %	7.2 %
B_RubAlKhali_megadunes	base	0.3	5.9 %	4.8 %
B_RubAlKhali_megadunes	gentle	1.0	13.4 %	9.5 %
B_RubAlKhali_megadunes	steep	1.0	22.4 %	11.9 %
B_RubAlKhali_megadunes	POST HOC: face within +/-1 post (3-post footprint)	1.0	32.2 %	23.1 %
B_RubAlKhali_megadunes	base, point sampling	1.0	10.9 %	9.3 %
C_RubAlKhali_linear	base	1.0	2.5 %	4.4 %
C_RubAlKhali_linear	base	0.6	0.8 %	2.7 %
C_RubAlKhali_linear	base	0.3	0.2 %	1.5 %
C_RubAlKhali_linear	gentle	1.0	3.5 %	3.4 %
C_RubAlKhali_linear	steep	1.0	3.7 %	6.4 %
C_RubAlKhali_linear	POST HOC: face within +/-1 post (3-post footprint)	1.0	22.0 %	14.4 %
C_RubAlKhali_linear	base, point sampling	1.0	2.1 %	4.3 %

Coverage versus certainty. Coverage is the share of cells that are not UNKNOWN. Certainty is the share of cells declared face free that are truly face free. On the synthetic ensemble (known truth) and on the real maps (model expected certainty, not a measured accuracy):

Posting, m	tau	Coverage (declared face free)	Certainty (truly face free among declared)	Missed face rate among declared
29.6	0.05	32.3 %	99.9 %	0.1 %
29.6	0.1	55.5 %	97.7 %	2.3 %
29.6	0.2	66.3 %	95.0 %	5.0 %
29.6	0.5	83.0 %	90.4 %	9.6 %
59.2	0.05	18.6 %	99.9 %	0.1 %
59.2	0.1	25.3 %	99.4 %	0.6 %
59.2	0.2	34.4 %	97.5 %	2.5 %
59.2	0.5	70.4 %	77.2 %	22.8 %
88.8	0.05	14.6 %	99.8 %	0.2 %
88.8	0.1	19.4 %	99.1 %	0.9 %
88.8	0.2	26.1 %	97.1 %	2.9 %
88.8	0.5	50.9 %	84.6 %	15.4 %

Area (30 m)	Face probability threshold tau	Coverage (not UNKNOWN)	Model-expected certainty of TRAVERSABLE cells (1 minus mean face probability)
A_south_Liwa	0.02	42.3 %	99.1 %
A_south_Liwa	0.05	67.3 %	98.2 %
A_south_Liwa	0.1	83.6 %	97.2 %
A_south_Liwa	0.2	91.4 %	96.2 %
A_south_Liwa	0.3	94.3 %	95.6 %
A_south_Liwa	0.5	97.0 %	94.6 %
A_south_Liwa	0.8	98.7 %	93.7 %
B_RubAlKhali_megadunes	0.02	37.8 %	99.1 %
B_RubAlKhali_megadunes	0.05	55.5 %	98.4 %
B_RubAlKhali_megadunes	0.1	82.4 %	96.4 %
B_RubAlKhali_megadunes	0.2	89.1 %	95.7 %
B_RubAlKhali_megadunes	0.3	91.8 %	95.1 %
B_RubAlKhali_megadunes	0.5	94.5 %	94.1 %
B_RubAlKhali_megadunes	0.8	96.4 %	93.0 %
C_RubAlKhali_linear	0.02	36.8 %	98.8 %
C_RubAlKhali_linear	0.05	75.5 %	97.8 %
C_RubAlKhali_linear	0.1	91.6 %	97.0 %
C_RubAlKhali_linear	0.2	97.5 %	96.3 %
C_RubAlKhali_linear	0.3	98.8 %	96.0 %
C_RubAlKhali_linear	0.5	99.7 %	95.7 %
C_RubAlKhali_linear	0.8	100.0 %	95.6 %

Detector skill on held out synthetic profiles. The detector separates cells with and without a face better than chance (AUC well above 0.5) and better than always predicting the base rate (Brier). It was also tested on ensembles with a different prior than the one it was trained on (gentle and steep dunes,

and a lower share of active faces), where the skill changes only moderately. **This is skill against the synthetic world it was trained on; it is not evidence about real dunes.**

Posting, m	Held-out synthetic ensemble (detector trained on base, pi 1.0)	Face rate	AUC	Brier	Brier of always predicting the base rate
29.6	base pi=1.0	22.2 %	0.924	0.086	0.173
29.6	gentle pi=1.0	13.7 %	0.954	0.057	0.118
29.6	steep pi=1.0	34.2 %	0.892	0.119	0.225
29.6	base pi=0.3	6.8 %	0.886	0.084	0.063
59.2	base pi=1.0	35.0 %	0.843	0.149	0.227
59.2	gentle pi=1.0	23.0 %	0.827	0.135	0.177
59.2	steep pi=1.0	49.0 %	0.846	0.161	0.250
59.2	base pi=0.3	10.7 %	0.778	0.151	0.095
88.8	base pi=1.0	44.7 %	0.857	0.151	0.247
88.8	gentle pi=1.0	31.6 %	0.794	0.170	0.216
88.8	steep pi=1.0	59.1 %	0.893	0.135	0.242
88.8	base pi=0.3	13.5 %	0.760	0.216	0.117

Effect on routes. Treating UNKNOWN cells as impassable, the share of the 10 origin destination pairs per window that stay reachable is:

Area	Posting, m	Pairs reachable (phase 1 rule, 10 pairs)	Pairs reachable with UNKNOWN impassable	Endpoints not TRAVERSABLE
A_south_Liwa	29.6	100 %	100 %	0 %
A_south_Liwa	59.2	100 %	70 %	20 %
A_south_Liwa	88.8	100 %	80 %	10 %
B_RubAlKhali_megadunes	29.7	100 %	100 %	0 %
B_RubAlKhali_megadunes	59.4	100 %	20 %	55 %
B_RubAlKhali_megadunes	89.1	100 %	40 %	45 %
C_RubAlKhali_linear	29.9	100 %	100 %	0 %
C_RubAlKhali_linear	59.7	100 %	80 %	10 %
C_RubAlKhali_linear	89.6	100 %	60 %	20 %

What this section establishes and does not. It establishes that, under stated assumptions about dune shape, a 30 m DEM leaves a modest share of cells (about a tenth in the Liwa and megadune windows, a few percent in the linear dune window) with a face probability above 0.20, that this share rises to roughly a quarter to a third if a face within one post counts, and that it is driven by the assumed morphology and the cell definition far more than by the four slip parameters. It does **not** establish that any cell is safe: a TRAVERSABLE cell has a face probability below 0.20, not zero. No claim is made that the detector would find real faces; the way to test it is to compare it with an RTK or drone profile of a real dune (the first step of `docs/NEXT_STEPS.md`).

---

9. Field-validation plan (PLANNED; nothing here has been run)

Full plan: `docs/FIELD_VALIDATION_PLAN.md`. Summary:

1. **Vehicle.** One small four-wheel rover (5 to 25 kg, wheels at least 150 mm, mass and wheel load known and kept fixed). Wheel diameter and width alter slip by large factors in the published lab data (LEG-017 to LEG-020), so the wheel must be fully described.
2. **Measurements.** Per-wheel encoders and motor current at 50 Hz or better, RTK GNSS ground truth (not rover odometry), a 6-axis IMU, rut depth by gauge at three stations per run, and tow tension if a load cell is available. Slip is computed against ground-truth speed with a calibrated effective radius, and distance-based slip is reported alongside because published studies use both.
3. **Soil, at every station.** Sieve grain size, bulk density, moisture at two depths, crust thickness, angle of repose, hand penetrometer profile and a plate pressure-sinkage set, with local slope from the RTK profile.
4. **Run design (about 60 runs).** Slope bins at 0, 5, 10, 15, 20 and 25 degrees (plus or minus 2), randomised order within a day, fresh track each run; a 12-run pilot to measure run-to-run scatter, then about 30 main runs; 9 cross-slope, 6 tethered descent and 3 repeated-pass runs. Run counts follow n = (1.96 sigma / E)^2: for sigma = 0.10 to 0.15 and E = 0.10 that is 4 to 9 runs per bin.
5. **Pre-registered analysis.** Calibrate the section 8 model on 0 to 10 degree runs only, **record dated predictions** for 15, 20 and 25 degrees before running them, then compare mean and root-mean-square error, the calibration slope and the entrapment slope, against a constant-slip baseline and a slope-threshold baseline. **Pre-data predictions are committed** (`docs/predictions/slope_ladder_predictions_2026-10-12.csv`, SIMULATED-BY-ME, from the assumption ranges only) and a script writes the post calibration predictions from the gentle slope runs before the steep runs. **Tolerance rule (assumption, fixed before the main block):** tolerance = the larger of 3 times the pilot sigma and 0.05, with pilot sigma the pooled within bin standard deviation of the 12 pilot runs; the model is called adequate if at least 80 % of held out runs fall within the tolerance and the entrapment slope is within 3 degrees; if 3 times the pilot sigma exceeds 0.30 the experiment is flagged uninformative. **Synthetic test with known truth (SIMULATED-BY-ME, `scripts/field_analysis_power.py`)** shows this rule discriminates weakly:

Synthetic truth	Noise sigma	Replicates	Called adequate	Failed	Flagged uninformative (3 x pilot sigma above 0.30)	Median tolerance
model	0.03	300	46 %	54 %	0 %	0.098
model	0.06	300	39 %	61 %	0 %	0.178
model	0.1	300	34 %	66 %	42 %	0.285
janosi_mean	0.03	300	24 %	76 %	0 %	0.114

Synthetic truth	Noise sigma	Replicates	Called adequate	Failed	Flagged uninformative (3 x pilot sigma above 0.30)	Median tolerance
janosi_mean	0.06	300	56 %	44 %	7 %	0.183
janosi_mean	0.1	300	78 %	22 %	43 %	0.286
collapse	0.03	300	31 %	69 %	20 %	0.208
collapse	0.06	300	35 %	65 %	34 %	0.255
collapse	0.1	300	38 %	62 %	64 %	0.340

The rule passes an exactly right model in under half of the replicates and fails a model that is wrong by construction (`collapse`: slip jumps to 0.85 from 22 degrees) in less than 70 %; at high noise it is lenient and flagged. **So an adequacy pass or fail from one ladder must not be over-read. Report the per bin predicted against observed slip with intervals and the entrapment slope as the primary result, and treat the pass or fail as secondary.** Table 9.1 shows what the protocol can and cannot predict before any data:

Table 9.1: pre-data slope ladder predictions (prior predictive from the assumption ranges, and the error band of the calibrate then predict protocol on synthetic truths).

Slope, deg	Role	Prior predictive slip, 5 to 95 % (phi 28 to 38)	Same with phi 33 plus or minus 2	P(slip at least 0.8), phi unknown	Protocol error band (predicted minus true), sigma 0.06
0	calibration	0.01 to 0.14 (median 0.05)	0.01 to 0.14	0 %	n/a (calibration slope)
5	calibration	0.03 to 0.23 (median 0.10)	0.03 to 0.22	0 %	n/a (calibration slope)
10	calibration	0.05 to 0.34 (median 0.15)	0.05 to 0.33	0 %	n/a (calibration slope)
15	held out prediction	0.07 to 0.53 (median 0.23)	0.07 to 0.49	1 %	-0.05 to +0.07
20	held out prediction	0.10 to 1.00 (median 0.36)	0.11 to 0.87	14 %	-0.08 to +0.41
25	held out prediction	0.16 to 1.00 (median 0.76)	0.18 to 1.00	48 %	-0.40 to +0.49

Calibrating on 0 to 10 degrees constrains the 15 degree prediction well, the 20 degree prediction weakly, and the 25 degree prediction hardly at all: the slip diverges near entrapment and gentle runs do not fix where that happens. The 25 degree bin is therefore mainly a test of the entrapment slope, not of the slip value. The same check found that calibration needs bounds on f_r and i0, because the two trade off at gentle slopes. 6. **Safety and scope.** Tether and recovery line, a second person at all times; slip faces near repose can avalanche. Not a safety case. Heat and dust are handled in `hudasol/dhill`. 7. **What to log for future prediction.** Wheel speeds and currents, RTK and IMU synchronised, effective radius, DEM-and-RTK slope and aspect, the soil state list, wheel state, ambient conditions, pass number and outcome flags.

The helper script `scripts/analyze_field_runs.py` and the log template `docs/field_log_template.csv` implement the analysis and logging; the script is tested on synthetic data only (including a wrong model case) and has never seen a field measurement. A one page checklist for the field day is `docs/FIELD_DAY_CHECKLIST.md`; low cost rover and measurement kit options, with costs only where a source was retrieved, are in `docs/ROVER_OPTIONS.md`; a proposal outline for a funded validation is `docs/PROPOSAL_SEED.md` (a proposal, not an achievement).

---

10. Limitations and honesty section

What this work cannot say without field data.

- It cannot say how well any model predicts slip, sinkage or entrapment of any rover on Arabian dune sand. No published number does either, and the simulation here is not evidence.
- It cannot say that any route is traversable. The model map is lower-bound optimistic (30 m grid) and its assumptions are unmeasured.
- It cannot say which model family (RFT, Bekker-Wong, SCM, DEM, learned) is best for dunes; no head-to-head test exists.
- It cannot give soil parameters for Gulf sand. The only grain sizes of natural dune sand retrieved are from Kuwait and Brazil.

Limits of the review.

- Retrieval was constrained to sources reachable from the sandbox and found by a US-only search tool. Paywalled journals and standard textbooks were not read; the per-topic notes list the exact URLs to fetch (`research/raw/*_notes.md`, sections NOT_RETRIEVED).
- Several values are secondary: tilt-bed slip numbers for a Mars-rover engineering model from a JPL press release (FLD-006, excluded from the comparison), a thesis (TM-022 to TM-024), conference slides (LEG-017 to LEG-020), and RoverDevKit's own re-evaluation of other authors' data (SIM-025 to SIM-027).
- Slip is defined differently across papers; several rows in the same figure measure different things. The warning column says how.
- Many simulation validations are plot-only. Rows with "qualitative" values carry no error.
- The rows' extraction and the first verification were done by language-model agents using the same search tools; shared blind spots (especially on figure-only values) are possible.
- UAE items are as of 2026-10-11 and some, such as the Rashid 2 launch date, conflict between sources.

Not done. Chrono, DEM and RFT were reviewed but not run. No dataset with slip labels from dunes was available, so no learning model was trained. The GLO-30 licence (COP-DEM-GLO-30-F) was read in full in phase 2 (`docs/DATA_AND_LICENSES.md`). Still not retrieved in full text: Wong and Reece 1967, Ishigami et al. 2007 and 2009, Heverly et al. 2013 and Takehana et al. 2025, and the Wong textbook parameter table; Yu et al. 2024 was abstract only. None is cited for any number. The sub-grid detector (section 8.7) has never been compared with a real lidar or drone profile of a slip face; the share of dune crests with an active face and the face height fraction of tall dunes are not published and are assumptions.

Licence. All rights reserved: the work is public for reading and review, and any use, modification, distribution or sale needs the author's prior permission (`LICENSE`, `LICENSE-DOCS.md`). Source PDFs and raw downloads are excluded by a CI check.

---

11. Verification

Evidence table, all 144 rows. Separate agents with no access to the extraction notes re-fetched sources. Phase 1 checked **30 rows drawn at random** (seed 20261011) from the 137 rows the extractors had not flagged, plus all 7 flagged rows; phase 2 checked the **remaining 107 rows** in four batches. Material error means a wrong value, unit, metric meaning, validation type, vehicle, terrain or slope; minor error means a correct number with a wrong label, section reference or descriptive detail.

Group	n	correct	minor	material	material rate	any-discrepancy rate
random sample, phase 1 (n=30 of 137 unflagged)	30	24	6	0	0 % (exact 95 % upper bound 9.5 %)	20.0 % (Wilson 95 % CI 9.5 to 37.3 %)
flagged sample, phase 1	7	4	3	0	0 % (exact 95 % upper bound 34.8 %)	42.9 % (Wilson 95 % CI 15.8 to 75.0 %)
remaining rows, phase 2 full re-check	107	80	21	6	5.6 % (Wilson 95 % CI 2.6 to 11.7 %)	25.2 % (Wilson 95 % CI 18.0 to 34.2 %)
ALL 144 ROWS	144	108	30	6	4.2 % (Wilson 95 % CI 1.9 to 8.8 %)	25.0 % (Wilson 95 % CI 18.6 to 32.7 %)

All errors were corrected through `research/corrections.csv` (old values in `research/dedup_log.md`). The six material errors found in phase 2: LRN-017 (a three wheeled prototype labelled four wheel), SIM-012 and LRN-006 (validation type: a prepared sand bed and the engineered JPL Mars Yard are `lab`, not `field`), LRN-019 (physical rover results only, `lab`, not `sim-vs-lab`), TM-020 (0.015 was a best case grid entry, the worst case is 0.070) and FLD-

029 (an unreferenced capability statement presented as a lab measurement; now validation type `none`). SIM-013 was set to `lab` for consistency with SIM-012 and carries the unit warning. The phase 1 sample had suggested zero material errors; the 107 rows showed that estimate was optimistic, which is why all rows were checked. Verifiers are language model agents using the same search tools, so shared blind spots (for example on figure only values) remain possible; a human spot check of a few rows is advisable. Duplicate rows between agents were removed (`research/dedup_log.md`); seven Kern et al. candidate rows read in phase 2 duplicate SIM-012 to SIM-014 and were not added.

Predictions register. Predictions were written and committed before each run (`docs/PREDICTIONS.md`). Hits and misses so far:

Stage	HIT	MIXED (part hit)	MISSED	pending
Stage 1	1	2	1	0
Stage 2	3	1	4	0
Stage 3	2	1	3	0

Report claims, second pass. A separate subagent that had not written the report checked the prose of this report (everything before Appendix A) against `research/evidence.csv`, `research/raw/*` and `results/*`, after the draft was complete. It examined 225 claim groups (every number in the simulation tables was compared individually and the statistics were recomputed): 188 correct, 26 minor (right number but a wrong label, row identifier or imprecise wording), 5 wrong and 6 unverifiable (WRONG rate 2.2 %; wrong plus minor 13.8 %). The five wrong items were a kc statement that belonged to another source (TM-029), percentage units applied to drawbar errors given in newtons (TM-001 to TM-004), a data-coverage cell that belonged to another product, a UAE vehicle date, and a tilt-bed row called a Mars measurement (FLD-006). **All 31 wrong and minor items were corrected in this text**; the six unverifiable statements were softened or removed. The full finding table is `research/factcheck_report_pass.md`. This checker is also a language-model agent working from the repository files, so it shares any error that is present in those files; it cannot catch a mistake that traces back to a source misread twice.

---

12. Open questions

1. **Sand variability.** How do grain size and shape distribution, packing, moisture and subsurface cementation (Liwa has sabkha interdunes with a shallow water table) change slip and sinkage, at one vehicle and slope? No quantitative Gulf data were found.

2. **Sim-to-field gaps.** Does a model calibrated on a flat or gentle slope predict slip at 15 to 25 degrees on the same dune sand? Is slope-induced shear collapse near repose important enough that a model must include avalanching?
3. **Resolution.** How far does a 30 m radar DEM understate the slope a rover feels at a slip face, measured against RTK profiles? Can aspect, curvature and dune height infer slip faces better than slope?
4. **Slip measurement.** Can a low-cost rover (RTK or total station, wheel encoders, IMU) give a slip estimate that agrees across the definitions used in the literature?
5. **Learning.** Does a model trained on non-sand terrain transfer to sand? Does a physical-prior hybrid generalise to Arabian dunes with a handful of onboard runs?
6. **Legged and tracked.** What are the same-vehicle differences on the same dune slope?
7. **Dynamics.** Crosswind, lateral slip on crests, speed and repeated passes.

Sources and appendices

Appendix A lists every evidence row. Appendix B lists the UAE facts used. Appendix C lists the sources, one line each, with the rows that cite them. All are generated from `research/evidence.csv` and `research/raw/uae_facts.csv` by `scripts/build_report.py`.

Appendix A. Evidence table (all PUBLISHED; condensed, full fields in `research/evidence.csv`)

144 rows. Validation types: field 38, lab 66, none 1, sim-vs-field 1, sim-vs-lab 18, simulation-only 20. Values are as printed in the source; read the comparability warning before comparing any two rows. Verification status of each row is in the CSV (column `verification`).

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
TM-001	Wheel C: aluminium cylinder coated with simulant, radius 130 mm, aspe...	Quikrete 1962 Medium Sand (MS), silica, 0.3-0.8 mm, loosely packed be...	0 (level)	RFT-granular	Drawbar mean absolute error (RFT vs experiment)	5.55 N	lab	link	Table 5 (MS, 80 N, RFT drawbar row, Delta); text...	2026-10-11	Forced-slip single-wheel rig on level bed; slip s=1-V/(omega*r) (Eq.15); error = mean absolute error over the tested slip values in range about -1..1, vs exper...
TM-002	Wheel C: aluminium cylinder coated with simulant, radius 130 mm, aspe...	Quikrete 1962 Medium Sand (MS), dry, loosely packed	0 (level)	RFT-granular	Drawbar mean absolute error (RFT vs experiment)	7.11 N	lab	link	Table 5 (MS, 190 N, RFT drawbar row, Delta)	2026-10-11	Forced-slip single-wheel rig on level bed; slip s=1-V/(omega*r) (Eq.15); error = mean absolute error over the tested slip values in range about -1..1, vs exper...
TM-003	Wheel C: aluminium cylinder coated with simulant, radius 130 mm, aspe...	Mars Mojave Simulant (MMS), crushed basalt, fines present, dry	0 (level)	RFT-granular	Drawbar mean absolute error (RFT vs experiment)	15.11 N	lab	link	Table 5 (MMS, 190 N, RFT drawbar row, Delta)	2026-10-11	Forced-slip single-wheel rig on level bed; slip s=1-V/(omega*r) (Eq.15); error = mean absolute error over the tested slip values in range about -1..1, vs exper...
TM-004	Wheel C: aluminium cylinder coated with simulant, radius 130 mm, aspe...	Mars Mojave Simulant (MMS), dry	0 (level)	RFT-granular	Sinkage mean absolute error (RFT vs experiment)	18.41 mm	lab	link	Table 5 (MMS, 190 N, RFT sinkage row, Delta); Sec...	2026-10-11	Forced-slip single-wheel rig on level bed; slip s=1-V/(omega*r) (Eq.15); error = mean absolute error over the tested slip values in range about -1..1, vs exper...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
TM-005	Wheel C smooth cylinder, radius 130 mm, vertical load 120 N	Poppy seeds (about 1 mm, NOT sand), dense (phi about 0.605) and loose...	0 (level)	classical-terrame...	Drawbar mean absolute error (TM, compacted)	3.41 N	lab	link	Table 3 (compacted, TM drawbar, Delta); Sec. 9.1...	2026-10-11	Poppy seeds not sand; TM parameters came from an intruder (2.5 × 3.8 cm2) much smaller than the wheel patch, and shear modulus was tuned; RFT drawbar error for...
TM-006	Wheel C smooth cylinder, radius 130 mm, vertical load 120 N	Poppy seeds (about 1 mm, NOT sand), dense state	0 (level)	classical-terrame...	Sinkage mean absolute error (TM, compacted)	26.07 mm	lab	link	Table 3 (compacted, TM sinkage, Delta)	2026-10-11	Poppy seeds not sand; see TM parameter caveat in previous row
TM-007	Pairs of cylindrical and four-arm lug wheels of different sizes, mass...	Quikrete medium sand bed, dry; sandpaper-covered cylindrical wheels	0 (level)	RFT-granular	Deviation of best-fit slope of scaled big-vs-small wheel da...	3 % (max, over six datasets)	lab	link	Experiments section, text after Fig. 4 (lug-wheel...	2026-10-11	Tests scaling invariance between wheel pairs, not absolute prediction error of slip/sinkage/drawbar; level terrain only; some deviation seen at highest rotatio...
TM-008	12-DOF hexapod, 4.04 kg, pointed dactyl feet (legged, free-walking)	Quikrete play sand in 15 cm deep sandbox, dry, settled for weeks	0 (level)	RFT-granular	5-step walking position error, no payload	6.84 %	sim-vs-lab	link	Sec. III-D (Speed and Sinkage of Walking Hexapod...	2026-10-11	Legged robot, not wheeled; level sandbox; single zeta tuned/matched to the box; sim tended to underestimate distance per step by 3.4-28%
TM-009	12-DOF hexapod, 4.04 kg plus payloads 0.5-2.277 kg	Quikrete play sand sandbox, dry	0 (level)	RFT-granular	Dactyl foot sinkage overestimate, mean across three payloads	2.6 %	sim-vs-lab	link	Sec. III-D, near Fig. 7E	2026-10-11	Experimental sinkage in Fig. 7E measured after a single trial per payload; foot sinkage not wheel sinkage
TM-010	12-DOF hexapod, 4.04 kg	Sandbox (Quikrete), Beach 1 (near shore), Beach 2 (farther from shore...	0 (level; beaches n...	RFT-granular	Average percent error of speed and body height (Beach 1 / B...	10.5 (Beach 1); 11.5 (Beach 2) %	sim-vs-field	link	Sec. III-E (Comparison with other Contact Models)...	2026-10-11	Beach grain properties not characterised beyond zeta; the sim used the sandbox zeta=3.75 although beach zeta was about 0.9; averaged speed+height error; legged...
TM-011	Bio-inspired L-leg and reversed L-leg rotated about hip (vertical-pla...	Poppy seeds (about 1 mm, NOT sand), loosely packed phi=0.58 via fluid...	0	RFT-granular	Vertical/horizontal force Fz,Fx vs leg angle, model vs meas...	within 15 %	lab	link	Sec. 3 text near Fig. 3c,d	2026-10-11	Poppy seeds not sand; leg not wheel; loose state only

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
TM-012	Small hexapedal robot (15 cm, 150 g) with L-legs	Poppy seeds (about 1 mm, NOT sand), loose	0	RFT-granular	Forward speed vs stride frequency	within 15 (L-legs); within 30 (reversed L-legs up to 3 Hz) %	sim-vs-lab	link	Sec. 4 text near Fig. 4b; leg-shape sweep within...	2026-10-11	Poppy seeds not sand; tiny legged robot; speed not slip
TM-013	Flat plate intruder on UR-10 arm (plate-level test)	0.8 mm glass beads (NOT sand), 18 cm deep bed	0	RFT-granular	Median relative force error /F1/ with beta-dependent alpha_...	1.2-6.5 %	lab	link	Sec. on plate validation text near Fig. 5 and Fig...	2026-10-11	Glass beads, plate not wheel; with constant alpha_y the error range is 1.9-34.1%
TM-014	Elliptical robotic foot driven by 2-link leg (human-gait replay)	Fine sand 0.2-0.5 mm (51×38×25 cm box with 20 cm depth was for plate...	0	RFT-granular	RMSE of vertical force Fz, slow gait (with correction vs wi...	21.4 +/- 19.5 (with); 25.8 +/- 18.9 (without) units not sta...	lab	link	Table I and text of Sec. IV-B (Robotic Foot-Terra...	2026-10-11	Foot intrusion, not wheel; unit of RMSE not printed in Table I; Fz error SD nearly equals the mean
TM-015	Smooth rigid wheel r=100 mm, b=75 mm, loads 14.9 / 47.7 / 63.9 N (tra...	Dry sand (donated by Shaw Brick) in 29 cm deep box, tests at slip 0.2...	0 (level)	classical-terrame...	Average error of mean of vertical force over all tests (as...	1.96 (SD 0.9) N	lab	link	Sec. 4 (Experimental Results), paragraph before F...	2026-10-11	Model calibrated on the 14.9 N case and eta tuned per case by eye; the paper focuses on oscillation amplitude/frequency (ripples), not mean slip/drawbar; copy...
TM-016	Grousered wheel (16 grousers, 23 mm), r=0.1 m, b=0.075 m, 15 N normal...	Dry sand (translated-terrain testbed)	0 (level)	classical-terrame...	RMS of mean-adjusted drawbar-pull oscillation: model 1.539...	0.536 units not stated in table	lab	link	Table 2 (Case Study #3, drawbar pull, slip 0.501...	2026-10-11	Metric is RMS of oscillation about the mean (mean-adjusted), NOT error in mean drawbar pull; units not printed; tuned on this very case
TM-017	Grousered wheel (32 grousers, each 10 mm long), same rig, 15 N	Dry loose sandy soil (translated-terrain testbed)	0 (level)	classical-terrame...	RMS drawbar oscillation: model 1.708 vs experiment 0.285 at...	1.423 units not stated in table	lab	link	Table 4 (Case Study #5, drawbar pull, slip 0.75 r...	2026-10-11	Same caveats as previous row; shows out-of-sample breakdown (model about 6x the measured oscillation)
TM-018	Rigid toothed-grouser metallic wheel, 250 mm dia, 112 mm wide, 12 gro...	RMCS14 olivine sand (quartz substitute), dry; DLR TROLL single-wheel...	0 (level)	other	Loss of drawbar pull coefficient at 20% slip, speed 0.01 to...	8 %	lab	link	Sec. 3.1 text near Fig. 5a	2026-10-11	Empirical measurement, not a model-vs-experiment error; included because quasi-static terramechanics/RFT assume speed independence; method_family set to 'other...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
TM-019	Rigid toothed-grouser metallic wheel, 250 mm dia, 112 mm wide, 12 gro...	RMCS14 olivine sand, dry	0 (level)	other	Loss of drawbar pull coefficient at 20% slip, speed 0.2 to...	63.7 %	lab	link	Sec. 3.1 text near Fig. 5a	2026-10-11	As previous row; gravity is 1 g; 100 N chosen to represent a roughly 350 kg six-wheel rover under lunar gravity
TM-020	Rigid grouser wheel D=150 mm, B=80 mm, 24 grousers of 8.5 mm, mass 10...	DEM model of FJS-1 lunar simulant (moderate cohesion), dry; coarse-gr...	-10 to +10 (slip ab...	DEM	RMSE of predicted vs direct lunar-g DEM slip ratio at theta...	0.015 slip ratio (dimensionless)	simulation-only	link	Sec. 4 text near Fig. 4a	2026-10-11	SIMULATION ONLY (DEM predicts DEM); no physical test; target is the scaling law, not DEM fidelity to a real soil
TM-021	Two grousered wheels, 10 kg wheel load, single-wheel testbed (wheel d...	Toyoura sand (source does not call it dry); bed 3500 × 600 x 1200 mm	0 in tests (slope h...	learning	Slip-ratio estimate 'accuracy', average (undefined in the p...	85 % (definition of accuracy not given in text read)	lab	link	Abstract and Sec. 4 text before Table 2; Table 1...	2026-10-11	In-sample goodness of fit, NOT a held-out prediction error (verifier arithmetic, not stated by the authors); transient slip-rate experiments
TM-022	Moonraker PFM2 rover about 7 kg, wheel radius 9 cm, width 8 cm, 15 gr...	Toyoura sand (spelled Toyura), 2.0 x 1.0 x 0.08 m tiltable sandbox; d...	10	classical-terrame...	Simulation error in side slip over travelled distance, base...	14.76 %	sim-vs-lab	link	Sec. 4.1.2, Table 4 (thesis p. 62)	2026-10-11	Thesis, not peer reviewed; soil parameters borrowed from another lab not measured in this sandbox; error is a ratio of side-slip difference to travelled distan...
TM-023	Moonraker PFM2 rover about 7 kg, wheel radius 9 cm, width 8 cm, 15 gr...	Toyoura sand, tiltable sandbox; dry not stated	20	classical-terrame...	Simulation error in side slip over travelled distance, base...	25.91 %	sim-vs-lab	link	Sec. 4.1.2, Table 4 (thesis p. 62)	2026-10-11	Thesis, not peer reviewed; soil parameters borrowed from another lab not measured in this sandbox; error is a ratio of side-slip difference to travelled distan...
TM-024	Moonraker PFM2 rover about 7 kg, wheel radius 9 cm, width 8 cm, 15 gr...	Toyoura sand, dry, tiltable sandbox	20	classical-terrame...	Simulation error in side slip over travelled distance, afte...	0.49 %	sim-vs-lab	link	Sec. 4.1.2, Table 4 (thesis p. 62)	2026-10-11	Thesis, not peer reviewed; soil parameters borrowed from another lab not measured in this sandbox; error is a ratio of side-slip difference to travelled distan...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
TM-029	Rectangular plates 25×75 to 70×140 mm (plate sinkage, not a wheel)	Loose dry Tottori sand, 350 mm deep bin, bulk density about 1.5 g/cm3	0	classical-terrame...	Goodness of fit (Wong's definition) of predicted pressure-s...	95 % goodness of fit	lab	link	Results section text on Fig. 4b; Table 2	2026-10-11	Plate sinkage only, small plates, shallow bin with bottom effect beyond 60 mm; goodness-of-fit is not a relative error
SIM-001	Single wheel on NASA Glenn TREC rig, 17.5 kg total load (half wheel,...	GRC-3 lunar simulant bed, cohesionless, flat, v=0.2 m/s, slip 0-0.8,...	0 (flat)	other	Drawbar pull (DBP) and traction slope vs slip, simulation v...	qualitative: described as good agreement; no numeric error...	sim-vs-lab	link	Sec 2.1 VV-mode single wheel experiments, pp.5-6...	2026-10-11	Simulant (GRC-3) not natural sand; agreement judged visually from plots; each point is one 20 s run; lab data from several repeats shown in figure only
SIM-002	MGRU3 (1/6-mass VIPER replica; 73 kg on GRC-3, 88 kg on GRC-1) full r...	GRC-1 and GRC-3 lunar simulants on tilted bed, wheel angular speed 0....	0 to 30 (steps of 5)	other	Slope-vs-slip (slope-mode) relationship, simulation vs SLOP...	qualitative: single-wheel and full-rover simulations match...	sim-vs-lab	link	Sec 2.2 slope-mode, pp.7-8 (Fig 7); count of 252...	2026-10-11	Slip defined from steady-state velocity at fixed angular velocity (slope-mode), not controlled slip; simulants not dune sand; rover is 1/6-mass replica tested...
SIM-003	VIPER (440 kg) simulated on Moon gravity	GRC-3 simulant properties, ramp 30 deg, lunar gravity (simulation onl...	30	other	Predicted wheel slip, VIPER on Moon	approx. 85 % slip	simulation-only	link	Sec 3 text discussing Fig 13, p.12	2026-10-11	Authors state Moon results could not be validated directly (no lunar test data; see their discussion near p.13); same text gives ~42% slip for the 73 kg MGRU3...
SIM-004	VIPER (440 kg) simulated on Moon gravity	GRC-3 simulant properties, ramp 10 deg, lunar gravity (simulation onl...	10	other	Predicted wheel slip, VIPER on Moon	approx. 75 % slip	simulation-only	link	Sec 3 text discussing Fig 13, pp.12-13	2026-10-11	As SIM-003; the same text gives ~18% slip for MGRU3 (73 kg) on Earth at 10 deg
SIM-005	Virtual bevameter plate-sinkage test: circular plates of 0.2 m and 0....	DEM soil bed generated by settling, DEM particle radius 0.005 m	0	multibody-SCM	Pressure-sinkage force: relative error of calibrated SCM vs...	<2 % of max force	simulation-only	link	Sec 3.1 (relative-error definition and result), p...	2026-10-11	SCM is compared to DEM, not to a physical bevameter; the authors say DEM fidelity versus a real bevameter has not been reported (Conclusions). DEM soil is not...
SIM-006	Annulus shear test rig, loads 25 to 200 kg	Same DEM bed as SIM-005	0	multibody-SCM	Steady-state shear torque: relative error of calibrated SCM...	<5 %	simulation-only	link	Sec 3.2 text near Figs 20-22, p.14-15	2026-10-11	Sim-vs-sim only

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
SIM-007	Single wheels: cylindrical (R 0.47 m, W 0.3 m), HMMWV wheel (R 0.47 m...	DEM bin 16 × 1 x 0.25 m with ~4.6 million particles; slip 0 to 0.8, v...	0	DEM	Wall-clock for 15 s single-wheel run: DEM vs SCM	DEM 4800 s; SCM 30 s (real-time factor about 320 vs 2, deri...	simulation-only	link	Sec 4.1, pp.16-17	2026-10-11	GPU model and SCM hardware: DEM on a GPU card (model not stated in the passage); DBP and traction-slope curves for SCM vs DEM described as comparing well, with...
SIM-008	Yutu-type six-wheel grouser rover, ~120 kg, wheelbase 1.136 m	DEM bin 14 x 2.5 × 0.375 m with ~15 million particles; v=1 m/s, slip...	0	DEM	Wall-clock for 10 s full-rover run: DEM vs SCM	DEM 10000 s (about 3 h); SCM 20 s (RTF about 1000 vs 2, der...	simulation-only	link	Sec 4.2, p.20; Conclusions	2026-10-11	SCM vs DEM agreement on DBP-slip and slope-slip stated qualitatively; DEM is itself unvalidated against a physical rover here
SIM-009	MGRU3 wheel (R 0.25 m, W 0.2 m, 24 grousers 25 mm high x 10 mm wide)...	GRC-1 simulant on ramp (gravity vector tilted), 5 x 0.8 × 0.25 m soil...	0 to 30	other	Wheel slip vs ramp angle: CRM vs experiment vs DEM	qualitative: close agreement with both lab data and publish...	sim-vs-lab	link	Sec 3.3, pp.9-10 (Fig 5a)	2026-10-11	Coarse SPH resolution (0.01 m) because of the 5 m domain; simulant, not dune sand; slip defined with grouser-inclusive effective radius
SIM-010	Cone penetration (not a wheel): 60 deg cone in glass beads and 30 deg...	Laboratory granular beds (glass beads; Ottawa sand), 4-inch Proctor-m...	0	other	Mean relative error of penetration depth vs experiment, ave...	glass beads 3.5; Ottawa sand 7.4 % mean relative error	sim-vs-lab	link	Sec 3.2 text and Fig 3 caption	2026-10-11	Not a wheel or rover test; error metric is penetration depth; experiments from a cited source (not retrieved), 3 repeats each
SIM-011	VIPER rover (1.24 M particles total, 0.28 M active) and M113 tracked...	Chrono CRM granular patches 4 × 2 × 0.1 m (VIPER) and 30 m S-path (tr...	0	other	Real-time factor (wall/sim; lower is faster)	VIPER: CRM 10.9 (optimised), SCM 50.7 default and 17.6 with...	simulation-only	link	Sec 4.3, Tables 3-4, p.21-22	2026-10-11	SCM grid set to 2x CRM particle spacing; authors call the CRM-vs-SCM comparison a rough estimate; RTF values are hardware-specific
SIM-012	EX1 four-wheeled grouser rover, 21.63 kg, 0.82 × 0.52 x 0.67 m, passi...	JAXA Sagamihara 20 x 20 m sand field, ~0.3 m deep dry loose Tohoku Si...	0	empirical-field	Steady-state slip ratio vs wheel speed (regression)	s = 0.0265*vw + 0.0256 (vw in m/s) slip ratio (fraction)	lab	link	Sec III-A and III-B, Eq. 2	2026-10-11	Indoor prepared sand field, not natural dunes; slip uses grouser-inclusive effective radius (excluding it gave negative slip); fit from 16 runs (as stated for...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
SIM-013	EX1 rover (21.63 kg) on slopes	Same Silica No.5 sand field, slopes 0 to 18 deg at speeds up to 0.47...	0 to 18 (fit claime...	empirical-field	Slip vs slope angle (regression term)	s(vw,alpha) = (0.00522*vw + 0.00105)*alpha^2 + s(vw) slip r...	lab	link	Sec III-B, Eq. 3	2026-10-11	Unit of alpha is not printed anywhere in the paper (full text read in phase 2). Degrees is inferred: with radians the slope term is about 0.0003 at 18 deg, so...
SIM-014	EX1 rover in OmniLRS / NVIDIA IsaacSim rigid-body model (Coulomb fric...	Lunaryard virtual terrain; regression slip and compliant-contact sink...	0 to 20	empirical-field	Simulation error relative to the authors' own regression: s...	slip <0.02 (flat); 0.22 mean absolute (slopes up to 20 deg)...	simulation-only	link	Sec IV (Results and Analysis, first paragraphs)	2026-10-11	The quoted errors are against the fitted model, not against independent field measurements; the field comparison (Fig 7, one run at 1.17 m/s) is described only...
SIM-015	Rashid-1 micro-rover wheel (R 100 mm, W 80 mm, 14 grousers 20 mm, 1.1...	Single-wheel testbed sandbox, Toyoura sand and FJS-1 simulant at ~25%...	0	DEM	Traction coefficient and dynamic sinkage vs slip, DEM vs si...	qualitative: good agreement (Figs 5-6); no scalar error rep...	sim-vs-lab	link	Abstract; Sec 3 first paragraph; Table 3	2026-10-11	DEM particles scaled 10-13x larger than real grains (2 mm); wheel is the closed-sided test wheel, not the flight design; fixed-slip testbed, not free-running r...
SIM-016	Same Rashid-1 wheel DEM (1.382 M particles)	Toyoura/FJS-1 virtual soil bin 0.65 × 0.14 x 0.14 m, 30 s run, dynami...	0	DEM	Wall-clock for a 30 s single-wheel run	about 7 h (RTF about 840, derived by me) h wall per 30 s sim	simulation-only	link	Sec 2.5 (Simulation Procedure) last paragraph	2026-10-11	Cost depends on frozen-particle dynamic domain; single GPU
SIM-017	Same Rashid-1 wheel DEM, same mass under Earth (24.5 N) and lunar (4....	As SIM-015, simulation-only extrapolation to lunar gravity	0	DEM	Simulated differences: lunar vs Earth gravity same mass; To...	Traction coefficient and tractive efficiency same Earth vs...	simulation-only	link	Abstract; Sec 3 (text before Table 4)	2026-10-11	No lunar-gravity validation data; simulation extrapolation only
SIM-018	Grouser wheel R 125 mm, W 100 mm, 12 straight grousers 25 mm, 80 N lo...	Toyoura standard sand in lab sandbox with observation window; Earth g...	not reported (flat...	DEM	Steady-state wheel slip ratio: simulation vs measurement	sim 12.1 vs lab 14.7 +/-3.0 (relative error about 18%); rut...	sim-vs-lab	link	Sec 4.1, Table 3 and following paragraphs; Table...	2026-10-11	2D single-layer DEM, free-slip run with controlled angular speed; friction angle used as tuning parameter; PIV window limits scale; error metric is relative er...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
SIM-019	Rigid DEM wheel D 71.27 cm, W 17.78 cm (wheel weight 1922 N in Table...	Dense sand, soil depth 20 cm in a 122 × 53 × 22 cm DEM box; lab cone...	0 (not stated)	DEM	Pull (drawbar) as % of wheel weight at 20% slip; gross trac...	Lab vs DEM: pull 17 vs 15; gross traction 31 vs 20; towed -...	sim-vs-lab	link	Sec 10 Table 2 and following text, p.11	2026-10-11	Authors' text calls the towed difference '6 percent' while the table shows 4 points; lab load in Fig 13 is 8230 N and rolling radius 41.3 cm versus 1922 N in T...
SIM-020	Same DEM rigid wheel	Same DEM setup (36 M particles)	0	DEM	Wall-clock to simulate 10 ms	26 h (RTF about 9 million, derived by me) h wall per 10 ms...	simulation-only	link	Sec 2 (DEM methodology), p.2-3	2026-10-11	2021 CPU-only cluster, fine particles; not comparable to GPU DEM rows
SIM-021	Underactuated tendon-driven soft gripper foot (D 250 mm), 5 to 66 N l...	Dry silica sand (close to Toyoura), 446 x 332 × 218 mm sandbox filled...	0 (comparison shown...	DEM	Pressure-sinkage curve: mean difference of simulated vs mea...	13.49 (SD 9.15) overall; 9.25 (SD 5.61) below 40 N % differ...	sim-vs-lab	link	Sec 3.2 and Fig 13	2026-10-11	Foot (not wheel); quasi-static load ramp over 10 s; particle clipping through gripper above ~50 N; sand not UAE dune sand
SIM-022	Same gripper foot, 66 N	Silica sand, sandbox tilted 0 deg	0	other	Maximum sinkage at 66 N	12.98 (5.2% of gripper diameter) mm	lab	link	Sec 2.3 (testing environment), Table 2; entry-spe...	2026-10-11	Slow manual load ramp; average of tests; entry speed changes result by ~20% on average (Sec 3.1)
SIM-023	Same gripper foot, 66 N	Silica sand, sandbox tilted 35 deg (about the angle of repose)	35	other	Maximum sinkage at 66 N	36.19 (14.5% of gripper diameter) mm	lab	link	Sec 3.1, Table 2	2026-10-11	Near angle of repose the material collapses; sinkage nearly triples vs flat; this lab row has no matching simulation
SIM-024	Same gripper EDEM+MotionSolve model	10 s, time step 0.01 s, 1.249 M particles	0	DEM	Wall-clock for a 10 s simulation	9 h (RTF about 3240, derived by me) h wall per 10 s sim	simulation-only	link	Sec 3.2, Table 3	2026-10-11	Hardware not stated in the passage I read
SIM-025	Rashid-1 micro-rover wheel (R 100 mm, 14 grousers, 24.5 N) - lab data...	FJS-1 simulant, single-wheel lab data	0	classical-terrame...	Median error vs measured drawbar pull and sinkage	DBP 24; sinkage 28 % median error	sim-vs-lab	link	Sec 5.1, p.17-18	2026-10-11	Compared to a published lab dataset (SIM-015 source), not own experiment; slip is an input; wheel load fixed; error metric is median per-point
SIM-026	Rigid smooth and grouser wheels, r=157 mm, 80 N (Ding et al. data)	Lunar simulant (full Bekker set reported by source)	0	classical-terrame...	Median error vs measured drawbar pull and sinkage	DBP 27; sinkage 52 (under-predicted) % median error	sim-vs-lab	link	Sec 5.1, p.18	2026-10-11	As SIM-025; sinkage error larger because rigid-wheel kernel under-predicts slip sinkage

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
SIM-027	Rigid wheel r=85 mm, 59 N (Wang and Han data)	KLS-1 simulant compacted to ~60% relative density; wheel sinks 1-14 mm	0	classical-terrame...	Median error vs measured drawbar pull and sinkage (stress c...	DBP 135 over-predicted; sinkage 385 over-predicted % median...	sim-vs-lab	link	Sec 5.1, p.18	2026-10-11	Firm, dense, lightly loaded regime outside the kernel's validity; author labels it a regime bound
SIM-028	Whole mission evaluation of a lunar micro-rover (<50 kg) including wh...	n/a (timings on one core of Apple M2 Pro, Python 3.12)	n/a	classical-terrame...	Computation cost	0.50 ms per wheel solve (median); 10.3 ms per mission evalu...	simulation-only	link	Abstract; Sec 3.2 'Soil parameters and runtime';...	2026-10-11	Single-machine timing, kernel is rigid-wheel and has the error bands in SIM-025 to SIM-027
SIM-029	RHex legged robot (27.4 kg) with 80 kg or 40 kg payload in Chrono	Digital twin of NASA Ames LHS-1 testbed (18 × 12 m grid, tamped/raked...	0 (straight path; t...	multibody-SCM	Qualitative outcome of traverse: immobilisation vs success	80 kg payload: about 100% slip and stop in medium-strength...	simulation-only	link	Sec 2.2.3 (simulation validation), p.9-10; Append...	2026-10-11	Authors call it 'validation' but it checks their risk-map model against SCM, with no physical measurement of slip; LHS-1 is lunar highland simulant not dune sa...
SIM-030	Archimede four-wheel-steer rover (~0.17 m diameter low-grouser wheels...	DLR PEL sandbox, granular lava 1-5 mm (Test Setup A, low-grouser whee...	0	other	End-point relative position error between odometric estimat...	sim <5; experiment <10 % of path	sim-vs-lab	link	Sec 4.1 and Conclusions (OCR of PDF; reported as...	2026-10-11	Metric is odometry error, not slip; no deformable terrain in simulation; the slope/slip results (5-20 deg) are experiment-only because no simulation was run fo...
LRN-001	KUSR (simulated)	Five heterogeneous terrain image samples (sample 4 = two lunar soils...	not reported	learning	MAE of slip-hazard estimate, samples 1-5	DeepLab+SlipNet 0.621/0.617/0.814/0.257/0.315; TerrainNet 0...	simulation-only	link	Table II (Sec. III Results)	2026-10-11	MAE is on a slip-risk scale, not a physical slip-ratio RMSE; absolute values of 0.6-0.8 on samples 1-3 are large and unit definition is ambiguous; not comparab...
LRN-002	KUSR (simulated)	Desert-sand-like terrain on path (a) of a heterogeneous test environm...	not reported	learning	predicted vs measured mean wheel slip ratio on desert-sand-...	predicted ~0.6; measured fluctuates (no summary number give...	simulation-only	link	Sec. III Results, discussion of Fig. 6 path (a)	2026-10-11	Qualitative read of a plot; simulation only; no error statistic reported for this path; large deviations attributed to sudden speed changes not modelled

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
LRN-003	LAGR research robot (2 front driven wheels, 2 rear castors) with ster...	Field sequences on soil, sand (flat), gravel (transverse slope), asph...	roll/pitch slopes f...	learning	slip-in-X prediction RMS, terrain type known: soil train/te...	soil 7.23%/11.8%; gravel 7.16%/27.5%; all datasets test app...	field	link	Sec. IV-B Results, Fig. 8 caption values and text...	2026-10-11	Test = later part of the same one-terrain sequence (temporal split), not held-out terrain; gravel test error high because roll/pitch combinations were absent f...
LRN-004	LAGR research robot	Composite field test set (~2000 frames) spanning soil, sand, gravel,...	not reported	learning	end-to-end slip prediction error (whole composite set); wit...	21.8% overall; 11.2% when terrain classified correctly % sl...	field	link	Sec. V-B Results, Fig. 10 and following text	2026-10-11	Test sequences stated not used for training, but terrain types (incl. sand) were seen in training; error inflated by misclassifying soil/gravel as sand with ~8...
LRN-005	LAGR research robot	Six-class patch classification (sand, soil, grass, gravel, asphalt, w...	not applicable	learning	terrain classification rate (pixels, 12×12 m map); per-clas...	76.4% overall; sand 0.85, soil 0.84, gravel 0.74, grass 0.3...	field	link	Sec. III, Fig. 7 (text labels extracted with pdft...	2026-10-11	Per-class numbers are extracted from a confusion-matrix figure, treat as approximate; test set ~30 frames per sequence, not held-out locations
LRN-006	Pluto research rover (JPL) with panoramic stereo imagery	JPL Mars Yard (55 m x 45 m): bedrock simulant, rocks, mixed sand/soil...	up to 30 (yard); pa...	learning	terrain classification accuracy, cell metric / pixel metric...	84.4% / 88.1% %	lab	link	Sec. 5 results, text after Fig. 12 (classificatio...	2026-10-11	Sand is merged into 'soil'; Mars Yard is a managed analog, test images from same yard as training (different perspectives); no numeric slip error reported in t...
LRN-007	MSL flight-spare wheel on MIT single-wheel testbed (IMU on wheel)	Mars regolith simulant with small ripples in the wheel path, ~20 m to...	not reported (level...	learning	slip-class accuracy, SVM (filtered) vs CNN (raw)	83% vs 88% % accuracy (3-class)	lab	link	Sec. 4.1 (15,000 samples, three slip classes) and...	2026-10-11	Classification into 3 bins, not continuous slip error; random hold-out (70/30, 60/40, 50/50) of one experiment series so test data are same soil and runs; no h...
LRN-008	Curiosity hazcam/navcam imagery and Fitorobot tracked robot imagery	11 image classes: 3 Mars classes (clean, rocks, boulders) plus gravel...	not applicable	learning	image terrain classification accuracy, CNN raw vs best case	87% vs 92% % accuracy (11-class)	field	link	Sec. 4.2-4.3 (300 + 800 images) and Sec. 5.2 text...	2026-10-11	Only 1100 images, random hold-out, mixed Mars flight imagery and Earth tracked-robot imagery; sand is one class among many

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
LRN-009	MSL flight-spare wheel, MIT single-wheel testbed (bin 3.14 m x 1.2 m...	Mars regolith simulant with ripples in wheel path; slip 0 to ~100%	not reported (level)	learning	training/test sample counts and class accuracy from regress...	9235 train / 3958 test; low and high slip classes ~85%, mod...	lab	link	Sec. 4.1 dataset split (9235/3958) and Sec. 5.2 t...	2026-10-11	Random 70/30 hold-out on one testbed dataset (in-distribution); numeric MAE only in Fig. 6 and not retrieved from text
LRN-010	Mars 2020 Perseverance rover (flight data); Earth-analog rover test b...	~500 sols, 45 km, ~43,000 images; rocky hills, flat bedrock, sand rip...	tilt logged per sam...	learning	hazard AUROC / AP / precision / recall / F1 on 35 held-out...	AUROC 0.874; AP 0.878; precision 0.835; recall 0.695; F1 0....	field	link	Table II and Sec. V Results	2026-10-11	Mars flight data; slip label = /wheel-odometry minus visual-odometry displacement/ over WO distance, hazard threshold at 90th percentile of slip, so binary cla...
LRN-011	Perseverance rover (flight data)	Held-out test sols including sand ripples, mixed rock and sand, a san...	not reported	learning	ablation hazard AUROC / quantile Spearman rho	full 0.874 / 0.721; no alignment 0.723 / 0.316; no Rank-N-C...	field	link	Table I and Sec. V-A	2026-10-11	As the main Perseverance row; Mars sand, not Earth dune sand
LRN-012	Curiosity, Opportunity, Spirit NAVCAM/MASTCAM imagery	Four classes: soil, bedrock, sand, big rock; ~35K images, ~326K crowd...	not applicable	learning	overall pixel accuracy on MSL NAVCAM expert test set; sand-...	94.97% (random-label training) and 96.67% (merged-label tra...	field	link	Tables 3 and 4 (3-label-agreement test set); abst...	2026-10-11	Terrain-type segmentation, not slip; test set is small (~1% of training set size) and drawn across orbital terrain classes; Mars flight imagery
LRN-013	Simulated vehicle in Verti-Bench (Chrono SCM-type deformable terrain)	Randomised deformable terrain regions with per-region soil parameter...	not reported	learning	32-step (3.2 s) rollout MAE x/y/z (m): NeSAM vs plain Trans...	0.80/0.78/0.06 m vs 0.89/0.90/0.09 m; yaw 4.18 vs 6.76 deg...	simulation-only	link	Table I and text below it (Sec. V-A)	2026-10-11	Trajectory-level kinodynamic error, not slip ratio; held-out 20% of trajectories from same simulator (in-distribution)
LRN-014	Verti-4-Wheeler (scaled 4-wheel vehicle), Verti-Arena	Grass, dirt and deformable sand regions with uneven geometry; referen...	not reported	learning	rollout MAE x/y/z; roll/pitch/yaw; closed-loop completion	0.068/0.059/0.012 m vs 0.082/0.070/0.019 m; yaw 5.79 vs 7.0...	lab	link	Table III, Table IV and Sec. V-B	2026-10-11	physical dataset is 50 trajectories of about 30 s (about 25 min, 15,000 transitions), 80/20 split by trajectory
LRN-015	Boston Dynamics Spot quadruped (joint position, velocity, effort)	Lava-tube floors (steep, sandy sloping floors in Mammoth Cave), limes...	described as steep/...	learning	slip prediction precision / recall / F-score (current time)...	95% / 80% / 0.84; 95% / 77% / 0.82 at 720 ms ahead %, F-sco...	field	link	Sec. IV Time-delay embedding and Future predictio...	2026-10-11	Binary slip-event detection, not slip magnitude; five-fold CV with 80/20 hold-out within pooled data from same sites (not terrain held out); sandy floor is onl...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
LRN-016	Simulated 10 kg rocker-bogie rover, wheel radius 0.4 m, width 0.1 m	Loose sand and compact sand per Wong equations	not reported (no sl...	learning	rover velocity prediction error (largest, linear model on l...	<5 % velocity error	simulation-only	link	Sec. 4 Experiments (text around Fig. 3-4)	2026-10-11	Held-out cohesion band 0.7-1.3 kPa and friction angle 20-45 deg were reserved for testing but data come from the same analytic Wong model used for training; no...
LRN-017	Three-wheeled rover prototype (two lugged driving wheels plus a follo...	Sand, stony sand, gravel, bedrock, rock; CTEM dataset 2375 images spl...	not reported	learning	segmentation accuracy (mIoU) and full-scale error of predic...	93.0% mIoU; 12.5% and 10.8% full-scale error; ground truth...	lab	link	Sec. V-A segmentation result and Sec. V-C text be...	2026-10-11	Predicts soil mechanical properties not slip; split by trial (better than frame split) but same yard; authors say result not accurate enough for control
LRN-018	Rover wheel with 3 piezoelectric sensors on hook-shaped spokes, drive...	Six terrains in total: flat, fine sand (<1 mm), small stone, small pe...	not reported (flat)	learning	terrain classification accuracy (1.5 s window, 120 random t...	90 % accuracy	lab	link	Sec. IV text around Fig. 11 (confusion matrix)	2026-10-11	Single speed, lab test platform; random trials not terrain-held-out; sand confused with large stones (attributed to slip on stones)
LRN-019	ERNEST 4-wheel rover with Active Gimbal Suspension (JPL)	M90 Mars soil simulant sand bed, 20 deg slope, dry and wet	20	learning	travel reduction (loss of expected no-slip distance) and co...	dry 81.9% vs 43.0%; wet 100% (entrapped) vs 30.0%; CoT dry...	lab	link	Table 6 and Sec. V text before it	2026-10-11	Control policy not slip prediction; travel reduction is slip-like but defined vs no-slip integrated motion; number of trials not stated; lab sand bed
LRN-020	Simulated cargo rover (20 kg + 0 to 80 kg payload), Project Chrono SCM	Single loose-soil SCM condition; grid of slope, heading, speed, mass	-15 to 10 (Table 1)	multibody-SCM	fit RMSE against SCM simulation: slip ratio and slip angle	0.0293 slip ratio; 0.0279 rad slip angle slip ratio; rad	simulation-only	link	Sec. 2 around Eq. 9-10 and Table 1/2	2026-10-11	RMSE is fit to its own training simulation (no held-out set); fitted parametric regressor, not a neural net; slopes only to 10 deg uphill
LRN-021	Single grousered wheel on carriage test bed (10 kg wheel load), bin 3...	Toyoura sand; constant slip-rate runs (slip rates +/-1, 2, 5, 10, 20...	not applicable for...	learning	adjusted R-square of per-slip-rate regression; mean slip-ra...	adj R2 0.695 to 0.973 (lowest at rate -20 and 1; highest at...	lab	link	Table 1 and Sec. 4 text; abstract	2026-10-11	In-sample curve fit, no held-out data; a regression not a deep model; RMSE column unit not stated so not recorded
LRN-022	Quasi-direct-drive quadruped, 2 gaits (Crawl N' Sense vs Trot-Walk),...	Damp construction sand at low/medium/high compaction; 300 micron beac...	0 (level)	other	crust rupture detection: Crawl gait specificity/sensitivity...	96%/63%; 12.5%/100% %	lab	link	Sec. V-B (Effect of gait design on detecting subs...	2026-10-11	Not learning-based; small sample (4 crawl, 3 trot trials); level artificial crust; shows gait choice changes proprioceptive reliability on sand

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
LRN-024	WVU Pathfinder, 15 kg four-wheel skid-steer	Mars-analog field site at Point Marion, PA; ~140-150 m runs (one ~650...	not reported	other	slip detection accuracy at ~150 m (abstract)	>92 % accuracy	field	link	Abstract; Sec. 5 Table 2 (confusion matrices not...	2026-10-11	Filter-based, not learned, and site is not a sand/dune site; medium and high slip bins often classified as low slip; accuracy degrades beyond ~500 m
FLD-001	MER Opportunity (6 wheels, ~26 cm wheel diameter, ~180 kg per Li et a...	Purgatory ripple, Meridiani Planum, Mars (sol 446); nondescript pile...	not reported	empirical-field	track length vs wheel-rotation distance during blind drive	2 m of track vs ~50 m of wheel travel m	field	link	Sec 5.1 text and Figure 10 caption (sol 446 Purga...	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Slip not computed by authors for this event; ratio is implied by the two lengths (about 96 percen...
FLD-002	MER Opportunity	Purgatory ripple escape drives, sols 463-483	not reported	empirical-field	VO-measured slip rate	98.9 to 99.5 %	field	link	Sec 5.1 paragraph beginning 'successful estimatio...	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Slip here is rover-level slip from visual odometry vs commanded/wheel-odometry distance, not sing...
FLD-003	MER Opportunity	Sandy slope near obstacle Wopmay, Endurance Crater, Mars (around sol...	17 to 23	empirical-field	slope range of sandy terrain where rover spent weeks failin...	17-23 deg	field	link	Figure 6 caption	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. No numeric slip given for this episode; slope is as stated by authors (local vs regional definiti...
FLD-004	MER Opportunity	Eagle Crater, loose soil and steep crater wall, Mars (crater diameter...	not reported	empirical-field	accumulated slippage over traverse	18.71 m slip over 91.42 m (20.5 %) %	field	link	Introduction paragraph [3]	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Cumulative slippage over >60 sols mixing slopes and soil types, not a per-slope value. Secondary...
FLD-005	MER Opportunity, first crater-slope attempt, Meridiani	Sandy crater slope Meridiani Planum, Mars (Feb 2004)	not reported	empirical-field	slip (wheel revolutions that did not produce progress)	50 %	field	link	Paragraphs on Lindemann's description of 4 wheel...	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Press-release description from JPL, not a peer-reviewed measurement; slope angle not stated for t...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
FLD-006	SSTB-lite (engineering model of MER, 'same weight and size, identical...	Variable-tilt platform with 6 inches of dry loose sand ('similar to c...	20 (uphill)	other	uphill slip at 20 degrees	90 cm of 100 cm commanded lost (10 cm progress) %	lab	link	'How Far Will The Rovers Slip and Slide' section	2026-10-11	Earth gravity lab tilt bed; release does not state whether rover mass was reduced to Mars weight. Slip defined by lost distance per commanded distance. Press r...
FLD-007	SSTB-lite	Same tilt platform, loose dry sand	25 (uphill limit)	other	uphill climbing limit (dead end) for Opportunity's sand	25 deg	lab	link	Baumgartner quote: dead-end point where rover can...	2026-10-11	Limit quoted by rover driver for 'the type of sand Opportunity is in'; derived from the tilt-platform plots, which were not retrieved. Earth gravity lab.
FLD-008	SSTB-lite	Same tilt platform, loose dry sand (downhill)	15 and 20 (downhill)	other	extra downslope slide while driving downhill	25 cm per m at 15 deg; 55 cm per m at 20 deg cm/m	lab	link	Same section, 'at a 15-degree angle facing down'	2026-10-11	Earth gravity lab tilt bed; negative-slip (skid) regime so not comparable to uphill slip percentages.
FLD-009	MER Spirit	Gusev Crater, Mars; slope >25 degrees (sol 206)	>25	empirical-field	maximum VO-measured slip	125 %	field	link	Sec 5 (introductory text before Sec 5.1), text ab...	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Slip here is rover-level slip from visual odometry vs commanded/wheel-odometry distance, not sing...
FLD-010	MER Spirit	Loose sand near 'Pot of Gold', Husband Hill base, Gusev, Mars	10 to 13	empirical-field	slip (reduction of distance traveled vs commanded)	up to 50 %	field	link	Sol ~200 section on repositioning at Pot of Gold	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Authors define slip as 1 - achieved/commanded distance (explicit in text).
FLD-011	MER Spirit	Columbia Hills, Gusev, Mars; appearance-identical terrain	19	empirical-field	slip	15 to 20 %	field	link	Paragraph 'extremely difficult to predict slip ba...	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Paired with the next row: PanCam imagery showed no visible difference between the two sites.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
FLD-012	MER Spirit	Columbia Hills, Gusev, Mars; ~30 m beyond the previous site, no visib...	16	empirical-field	slip	95 %	field	link	Same paragraph	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Shows slip not predictable from slope or imagery alone: 95 percent at 16 deg vs 15-20 percent at...
FLD-013	MER Spirit	Clovis outcrop approach, Husband Hill, Gusev, Mars	up to 30 (tilt)	empirical-field	slip in one case	over 100 %	field	link	Paragraph on Clovis (sol ~203 onward)	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Value is a single reported case with tilt up to 30 deg; terrain mix of rock and soil, not pure sa...
FLD-014	MER Spirit	Husband Hill traverse, Gusev, Mars; 'fine grain' class (short VO segm...	10 to 15 (onset)	empirical-field	longitudinal slippage on fine-grain terrain	approaches 100 at slopes as early as 10-15 deg %	field	link	Sec 5.2 paragraph [40], Figure 11a	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Terrain class assigned from images; slope from local DEM/segment fit; slippage defined per segmen...
FLD-015	MER Spirit	Husband Hill traverse, long segments, sand-sized class	0 to 5	empirical-field	relative longitudinal slippage	4.0 (49 segments, 528.56 m) %	field	link	Table 6 (sand-sized rows)	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Segment-averaged over mixed conditions; authors note higher slopes sometimes show smaller slippag...
FLD-016	MER Spirit	Husband Hill traverse, long segments, sand-sized class	5 to 10	empirical-field	relative longitudinal slippage	9.9 (36 segments, 239.49 m) %	field	link	Table 6 (sand-sized rows)	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Same caveats as previous row.
FLD-017	MER Spirit	Husband Hill traverse, long segments, fine-grain class	5 to 10	empirical-field	relative longitudinal slippage	16.6 (9 segments, 61.78 m) %	field	link	Table 6 (fine grain rows)	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Few segments; no fine-grain drives above 15 deg (authors).

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
FLD-018	MSL Curiosity (6 wheels; ~0.5 m wheel diameter per Li and Lewis 2022...	Megaripple field, Gale Crater, Mars (sol 1181 controlled crossing)	slightly uphill (va...	empirical-field	rover 3D slip over ~8 m of shallow megaripple sand	exceeded 70 in places %	field	link	Left column bullets, sol 1181-1185 crossing exper...	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Slip here is rover-level slip from visual odometry vs commanded/wheel-odometry distance, not sing...
FLD-019	MSL Curiosity	Uphill sandy terrain, Gale Crater, Mars (sol 1703 drive)	9.5 (rover pitch)	empirical-field	slip at fault	61.2 (after 2.88 m) %	field	link	Flight data-set comparison, Figure 25 (after the...	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Pitch is the rover body pitch, not a terrain-slope measurement; slip method presumably VO-correct...
FLD-020	MSL Curiosity	Gale Crater traverses (flight software default)	n/a	other	default autonomous-drive slip limit	60 %	field	link	Same passage: 'default 60% slip limit'	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Operational limit, not a measurement.
FLD-021	MSL Curiosity	Gale Crater traverses (as cited by Kilic et al., secondary)	n/a	other	stop thresholds: single slip measurement; consecutive slip...	single 70-90 percent; consecutive 40-70 percent %	field	link	Slip ratio discussion in Sec 3.1	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Secondary citation of Arvidson et al. 2017 (full text not retrieved); expressed in the source as...
FLD-022	MSL Curiosity with Traction Control enabled vs disabled	Sandy/rocky drives sols 1664-1707, Gale Crater, Mars	not reported (terra...	empirical-field	change in average max rover slip with Traction Control	2.1 % lower with enabled (absolute vs relative not specifie...	field	link	Discussion of Figure 25	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Small sample, not randomised terrain; the sol 1703 excessive-slip sand drive was removed from the...
FLD-023	Perseverance ENav tilt-limit setting, derived from Scarecrow (Curiosi...	Jezero planning limit referencing Curiosity-class Scarecrow test in p...	16 (max allowed til...	other	max terrain tilt allowed for Perseverance autonomous nav (a...	16 deg	field	link	'Flight Software Update' section (text partly gar...	2026-10-11	Design limit justified by Earth dune test with the Scarecrow rover (cited to Heverly 2013, not retrieved) and then applied to Mars; not a measured slip or sink...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
FLD-024	Perseverance (Mars 2020)	Summerland Trail, Jezero crater rim hill, Mars (sols 1283-1305)	15 to 20	empirical-field	autonav slip limit needed (Slip Tables)	above 50 %	field	link	Mobility Challenges / Slip Issues section	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Slip here is rover-level slip from visual odometry vs commanded/wheel-odometry distance, not sing...
FLD-025	Perseverance (Mars 2020)	Midtoya, west Jezero crater rim, very sandy terrain with small loose...	about 20	empirical-field	slip	in excess of 94 %	field	link	Mobility Mechanism Issues section	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Slip here is rover-level slip from visual odometry vs commanded/wheel-odometry distance, not sing...
FLD-026	Perseverance (Mars 2020)	Small sand ripples near preferred path, Jezero crater floor, Mars (Ra...	not reported	empirical-field	wheel slip over small ripples	not quantified; described as insignificant n/a	field	link	Section 6/7 text around 'sand ripples near the pr...	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. No numbers; shows small/low ripples on firm substrate are not equivalent to megaripples or dune f...
FLD-027	Scarab rover (400 kg, 4 rigid wheels 71 cm x 18 cm, sandpaper-covered...	NASA Glenn SLOPE lab tilt bed, 6 × 4.5 m, 0.23 m deep GRC-1 lunar reg...	20	other	longitudinal slip at 90 and 75 deg angle of attack	exceeded 90 %	lab	link	Sec 3.4.1 (Figures 3.6, text after Fig 3.5)	2026-10-11	Lunar simulant, NOT dune sand; Earth gravity with a heavy rover; rigid wheels without grousers. Slip definition is velocity-based at constant wheel speed 1.24...
FLD-028	Scarab rover (same)	SLOPE lab tilt bed, GRC-1 simulant	25	other	longitudinal slip at 90 deg angle of attack (direct ascent)	100 (no angle of attack made uphill progress) %	lab	link	Sec 3.4.1, Figure 3.7 discussion	2026-10-11	Same as previous row.
FLD-029	Andy 2 (small 4-wheel skid-steer lunar rover)	loose soil (not tied to GRC-1 for this claim)	15	other	slip when ascending	20 to 30 %	none	link	Sec 2.2.2 (subsection Experiment setup and proced...	2026-10-11	Unreferenced author capability statement in Sec 2.2.2; not a measurement, not tied to GRC-1; do not use as a comparator
FLD-030	El-Dorado-II (35 kg, 4 wheels each 178 mm dia x 100 mm wide, no grous...	Tiltable sandbox 2 × 1 m filled with dry Toyoura standard sand, Earth...	4	other	slip ratio, direct uphill ascent	close to 0.9 (text); read from figure description ratio	lab	link	Appendix B.2.2	2026-10-11	Light rover, small wheels; Toyoura sand is uniform fine sand, not natural dune sand. Slip ratio velocity-based.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
FLD-031	El-Dorado-II with 12 grousers (15 mm)	Same Toyoura sand slope bed	4	other	slip ratio, direct uphill ascent	around 0.2 ratio	lab	link	Appendix B.2.2	2026-10-11	Same as previous row. More grousers lowered slip further but less so above 10 deg.
FLD-032	Four-wheel rover, 2 driven front wheels, 23 kg, wheel 296 mm dia x 90...	Indoor sandbox, 0.725 m3 dry Toyoura sand 250 mm deep, incline 0-30 d...	10	other	maximum sinkage angle, 20 mm grouser	127 deg	lab	link	Section 4.1 (Figure 12b)	2026-10-11	Only front wheels driven; sinkage reported as an angle subtended at the wheel centre, not mm; slip ratio plotted but not stated in text.
FLD-033	Same rover, 80 mm fixed grousers	Same sandbox	10	other	maximum sinkage angle	66 deg	lab	link	Section 4.1 (Figure 12b)	2026-10-11	Same as previous.
FLD-034	Same rover, 80 mm fixed grousers	Same sandbox	20	other	maximum sinkage angle (smallest across grouser lengths)	118 deg	lab	link	Section 4.1 (Figure 12c)	2026-10-11	Same as previous. Authors judge >120 deg as stuck; 20 A current limit stopped runs near that.
FLD-035	Same rover, all grouser lengths 20-80 mm	Same sandbox	30 (at angle of rep...	other	forward progress per wheel revolution at slope = angle of r...	less than 10 mm (wheels spun in place) mm/rev	lab	link	Section 4.1 (Figure 12d discussion)	2026-10-11	Same as previous.
FLD-036	SpaceBok quadruped (legged), planar vs point feet	RUAG Space ExoMars locomotion testbed 6 × 6 m, 20 t ES-3 Mars soil si...	0 to 25	other	max foot sinkage with point feet	up to 120 mm	lab	link	Section 4.2.1	2026-10-11	Legged robot, simulant, no gravity compensation; qualitative ascent table.
FLD-037	Hexapod robot (350 g, 22 × 10 cm, C-legs)	Tiltable fluidised bed of 300 micron glass beads, 127 × 65 cm, Earth...	0	other	step-averaged speed on level bed	15.7 +/- 0.97 cm/s	lab	link	Section III-A	2026-10-11	Glass beads not sand; small legged robot; speed not a slip ratio.
FLD-038	Hexapod robot (same)	Same tilt bed	10 (11.0 +/- 0.14 c...	other	step-averaged speed at 20 deg	4.98 +/- 0.32 cm/s	lab	link	Section III-A	2026-10-11	Same as previous.
FLD-039	SSTB-Lite (MER model) stripped to Mars-weight; Scarecrow (Curiosity-c...	Dumont Dunes OHV Area near Baker, California (Earth dunes), May 2012	not reported (dune...	other	uphill vs downhill ground speed at equal wheel rate	uphill much slower than downhill (qualitative) n/a	field	link	Article body	2026-10-11	Qualitative blog/press statement; the uphill-slower-than-downhill observation is for SSTB-Lite only, no slope or slip numbers. Mars-weight vehicles at Earth gr...
FLD-040	Opportunity (MER)	Purgatory ripple, Mars	not reported	other	ripple height and entrapment duration	30 cm high; several weeks embedded cm	field	link	Figure 1 caption	2026-10-11	Mars data are field data from Mars (0.38 g), NOT Earth dunes. Review/perspective source, descriptive.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
FLD-041	ESA/DFKI rover field campaign (Mana, Minnie, SherpaTT)	Sahara dunes near Erfoud, Morocco (Dec 2018 field test)	not reported	other	vision-feature tracking on smooth dunes	difficult; algorithms behaved unexpectedly (qualitative) n/a	field	link	Quote paragraph by G. Visentin	2026-10-11	ESA news item; no slip, sinkage or slope numbers. Relevant only to the failure of VO on featureless dune faces.
FLD-042	Husky UGV (Graz University of Technology) in AMADEE-18	Dhofar desert base camp area, Oman (Feb 2018); loose soft sand	not reported	other	stuck events and dead-reckoning error	rover got stuck from time to time; dead-reckoning error lar...	field	link	Section 10.8 Summary	2026-10-11	Only Arabian wheeled-robot field evidence found; no slip, sinkage or slope numbers. Robot was a proof-of-concept mapping rover, not a mobility study.
LEG-001	SandBot RHex-class hexapod (total mass 2.3 kg; c-legs R=3.55 cm, hip...	Lab fluidized-bed trackway of ~1 mm poppy seeds, volume fraction 0.60...	0 (level)	RFT-granular	mean forward speed	2 cm/s	lab	link	Sec. 3 (Methods and Results), Fig. 2c caption (HG...	2026-10-11	Poppy seeds in fluidized bed, 2.3 kg tethered-power hexapod; scale, grain size (1 mm) and packing control are not representative of dune sand; speed not normal...
LEG-002	SandBot RHex-class hexapod (2.3 kg), soft-ground clock (SGK)	Lab fluidized-bed trackway of ~1 mm poppy seeds, volume fraction 0.605	0 (level)	RFT-granular	mean forward speed	8 cm/s	lab	link	Fig. 2c caption (SGK ~8 cm/s); Sec. 3 states opti...	2026-10-11	As LEG-001; gait parameters were hand-tuned to this substrate and packing.
LEG-003	SandBot hexapod (2.3 kg), tuned rotary-walking gait, leg frequency 16...	Lab fluidized-bed trackway of ~1 mm poppy seeds, volume fraction 0.633	0 (level)	other	mean forward speed	20 cm/s	lab	link	Results, text around Fig. 1d,e (vx ~20 cm/s at ph...	2026-10-11	Model granular medium; packing fraction control is not available in dune sand; hexapod mass 2.3 kg.
LEG-004	SandBot hexapod (2.3 kg), same gait, leg frequency 16 rad/s, loosely...	Lab fluidized-bed trackway of ~1 mm poppy seeds, volume fraction 0.600	0 (level)	other	mean forward speed	2 cm/s	lab	link	Results, text around Fig. 1d,e (vx ~2 cm/s at phi...	2026-10-11	As LEG-003. A 5% change in packing fraction changed speed ~5-10x: highly packing-sensitive.
LEG-005	DASH hexapod (22.1 g, ~10 cm), sprawled posture	Lab fluidized-bed trackway of poppy seeds, loosely packed, volume fra...	0 (level)	other	mean forward speed (normalized)	5 body lengths/s	lab	link	Sec. 4 Results, paragraph after Fig. 7 (about 5 b...	2026-10-11	~22 g, 10 cm robot; thrust-to-weight far higher than any field vehicle; poppy seeds not dune sand; BL/s not comparable to m/s of larger vehicles.
LEG-006	RoACH hexapod (24.9 g, ~10 cm), upright compliant legs	Lab fluidized-bed trackway of poppy seeds, loosely packed, volume fra...	0 (level)	other	mean forward speed	5 cm/s	lab	link	Sec. 4 Results (RoACH loses >90% of performance,...	2026-10-11	~25 g robot; poppy seeds not dune sand.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
LEG-007	Hexapod with C-shaped legs (350 g, 22 cm x 10 cm; leg radius 4 cm, wi...	Lab tiltable fluidized bed of 300 um glass beads, loosely packed, dep...	0	RFT-granular	step-averaged speed parallel to surface	15.7 cm/s	lab	link	Sec. III-A (v = 15.7 +/- 0.97 cm/s at 0 deg)	2026-10-11	350 g robot on glass beads in 127×65 cm bed; slope is bed tilt of uniformly prepared bed, not natural dune with variable compaction; slope max 24 deg due to an...
LEG-008	Same 350 g hexapod	Same tiltable glass-bead bed (300 um)	10	RFT-granular	step-averaged speed parallel to surface	11.0 cm/s	lab	link	Sec. III-A (v = 11.0 +/- 0.14 cm/s at 10 deg; 10....	2026-10-11	As LEG-007.
LEG-009	Same 350 g hexapod	Same tiltable glass-bead bed (300 um), near angle of repose	24	RFT-granular	step-averaged speed parallel to surface	5.6 cm/s	lab	link	Sec. III-A (v = 5.6 +/- 1.6 cm/s at 24 deg; 4.98...	2026-10-11	As LEG-007; 24 deg is at the bed's angle of repose, not extrapolable to dunes.
LEG-010	SpaceBok quadruped (22 kg, hip height 0.5 m), point feet	Lab Mars-analog soil testbed, 6×6 m, 20 t of ES-3 (coarse, poorly-sor...	0-25 (ascent/descen...	other	maximum foot sinkage observed	120 mm	lab	link	Sec. 4.2.1 (point feet sank up to 120 mm into the...	2026-10-11	Simulant ES-3 is a Mars-analog soil in a lab bin, not aeolian dune sand; 22 kg legged robot with hand-tuned gaits; sinkage reported qualitatively as a maximum.
LEG-011	SpaceBok planar adaptive foot, 110 cm2, with 12 mm grousers (single-f...	Lab single-foot traction test on ES-3 simulant	not reported for tr...	other	traction coefficient increase with grousers vs same foot wi...	22 percent	lab	link	Sec. 3.3.2 (traction coefficient raised by 27% on...	2026-10-11	Foot-pad grousers on a legged foot, not wheel grousers; single-foot rig, not full vehicle.
LEG-012	SpaceBok quadruped (22 kg), dynamic (trot) vs static walk, planar and...	Lab Mars-analog ES-3 slope testbed	0-25	other	energy per distance, trot relative to static walk	>2 (trot more efficient by more than 2x) ratio	lab	link	Sec. 4.2.3, text before Fig. 15 (dynamic gait mor...	2026-10-11	Ratio only; absolute cost of transport not extracted. Hand-tuned gaits; lab bin.
LEG-013	SpaceBok quadruped (22 kg), planar and point feet, static and trot	Lab Mars-analog ES-3 slope testbed (max tilt 25 deg)	25	other	maximum slope traversed (testbed limit)	25 deg	lab	link	Sec. 4.2.1 and Table 2 (up to 20 deg no significa...	2026-10-11	25 deg is the testbed maximum, not a failure limit. Qualitative success scoring (pie chart), numerical success rate not reported. Slope was a bin tilt of prepa...

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
LEG-014	Single-leg hopper (2 kg robot, direct-drive) with active damping cont...	Lab sandbox of 3.4 mm glass beads, 0.16 m deep, controlled compaction	0 (vertical jump)	other	reduction in energy per jump vs compression-extension contr...	15 percent (10-15 range across foot size and stiffness)	lab	link	Sec. 3.2 / Table 4 (savings 14, 15, 10, 11% acros...	2026-10-11	Single hopping leg on rail in glass beads, jump energy not horizontal cost of transport; 3.4 mm glass beads not dune sand.
LEG-015	Unitree G1 humanoid (hardware), RFT-trained RL policy vs rigid-contac...	Beach volleyball court dry sand and basalt, outdoor hardware tests	0 (level; slope not...	learning	highest commanded speed tested (2.5 m/s) at which RFT-train...	2.5 m/s	field	link	Sec. V-B (Results, Hardware Experiments), Fig. 7	2026-10-11	Humanoid biped, commanded not measured speed; success defined as staying upright without human help; sand untested for slope; numeric success rates not extract...
LEG-016	Four-module wheel-paddle (wheel-leg hybrid) robot, mass 24 kg, hybrid...	Outdoor sand field at Ritsumeikan University BKC, Shiga, Japan (Sand...	0 (level; slope not...	other	traversing speed approaching 1 body length/s when wheel joi...	1 body lengths/s (approx.)	field	link	Sec. 4.4.2 text around Fig. 4.15; sand-least-effi...	2026-10-11	Single tested site, inhomogeneous sand, body length value not extracted, 24 kg robot; wheel-leg hybrid with protruding paddles (not wheel-only).
LEG-017	Four inline rigid lugged wheels, D=116 mm (lug 5 mm, 24 lugs, width 1...	Lab sand box of Toyoura sand	12 (equivalent slop...	other	slip ratio	0.741 -	lab	link	Slide 38 ('Wheels with different dimensions' and...	2026-10-11	Slope is emulated by horizontal pull, not an actual incline; wheel vertical load not stated on retrieved slides; slides, not peer-reviewed paper.
LEG-018	Four inline rigid lugged wheels, D=327 mm (lug 15 mm, 24 lugs, width...	Lab sand box of Toyoura sand	12 (equivalent slop...	other	slip ratio	0.056 -	lab	link	Slide 38	2026-10-11	As LEG-017. Diameter and lug height varied together, so effects are not separable.
LEG-019	Four inline rigid lugged wheels, D=202 mm, width 50 mm	Lab sand box of Toyoura sand	14 (equivalent slop...	other	slip ratio	0.471 -	lab	link	Slide 40 (width series at D=200 mm)	2026-10-11	As LEG-017; slope emulated.
LEG-020	Four inline rigid lugged wheels, D=202 mm, width 150 mm	Lab sand box of Toyoura sand	14 (equivalent slop...	other	slip ratio	0.247 -	lab	link	Slide 40	2026-10-11	As LEG-017; slope emulated.
LEG-021	Rigid grousered wheel, radius 114 mm, width 57 mm, 13 mm grousers, 20...	Lab glass-walled soil bin with GRC-1 lunar simulant (not dune sand)	0 (level)	classical-terrame...	number of grousers beyond which drawbar pull stops improving	24 count	lab	link	Sec. IV and Fig. 11, Table II (13 mm: 24 predicte...	2026-10-11	Lunar simulant, single wheel at fixed 20% slip, wheel about MER scale.

ID	Vehicle or wheel	Terrain	Slope, deg	Method family	Metric	Value	Validation	Source	Location	Retrieved	Comparability warning
LEG-022	Flexible metallic grousered wheel vs rigid wheel, both D=250 mm, W=11...	Lab soil bin with olivine sand RMCS14	0 (level)	other	drawbar pull and tractive efficiency of flexible vs rigid w...	>2 (over two times higher) ratio	lab	link	Abstract and Sec. 4 discussion (flexible wheel at...	2026-10-11	Single-wheel bin test at 20% slip, 100 N load; flexible wheel is a metallic rover wheel not a non-pneumatic tire for a heavy vehicle.
LEG-023	OriWheelBot origami-wheeled robot (1348.5 g), wheel width 22 mm vs 72...	Lab soft sand bed with subsidence pit (starts 30 mm sunk)	0 (level)	other	distance travelled in 8 s after starting sunk in sand, narr...	60 mm	lab	link	Sec. on soft sand walking (dpt about 60, 130, 330...	2026-10-11	Small 1.3 kg robot with deformable origami wheel; sand uncharacterized; start-from-sunk test, not steady traversal.
LEG-024	OriWheelBot (1348.5 g), wheel width 72 mm vs 22/38/52 mm	Lab sand slope, about 17 deg	17	other	climbs 17 deg sand slope (success only at widest wheel)	success at 72 mm; failure at 22, 38, 52 mm pass/fail	lab	link	Sec. on slope climbing, Fig. 5b caption	2026-10-11	One slope angle, one trial class, binary result; sand uncharacterized.

Appendix B. UAE facts (from `research/raw/uae_facts.csv`)

ID	Fact	Source	Status
UAE01	Rashid 1 rover mass was about 10 kg.	link	verified-from-source
UAE02	Rashid 1 had four wheels; wheels designed to climb slopes up to 20 degrees and rocks up to 10 cm high.	link	verified-from-source
UAE03	Rashid 1 instruments: two wide-field navigation cameras, a microscopic imager and a thermal imager (Langmuir probe also listed by a secondary source).	link	partially-verified
UAE04	A UAE wheel-terrain paper gives the Rashid wheel as 0.1 m diameter, 7.5 mm wide, 12 fixed grousers 10 mm high, wheel weight 1 kg.	link	verified-from-source
UAE05	The same paper modelled soil with a modified Reece pressure-sinkage model plus Wong shear model, using an MBRSC regolith sample representing Lacus Somniorum; simulator was Vortex Studio.	link	verified-from-source
UAE06	Single-wheel rig results at slip 0, 0.25, 0.5, 0.75: experimental sinkage 0.010, 0.0158, 0.0169, 0.0192 m; drawbar pull 5.2, 10.1, 16.77, 24.1 N.	link	verified-from-source
UAE07	Khalifa University master's thesis (2022, Mariam Alhammadi) analysed Rashid-1 wheels with terramechanics and FEA; reports peak tractive efficiency at 10-15% slip and that larger grousers raise drawbar pull while general...	link	verified-from-source
UAE08	A 2022 IAC paper (Alhammadi et al.) used finite element analysis on Rashid wheel and grouser geometry on lunar regolith; beyond a point larger diameter, more or taller grousers do not further reduce required drawbar pul...	link	verified-from-source
UAE09	Samples from ESA, CREST (Universite libre de Bruxelles), Cambridge Graphene Centre, York University and others were bonded to Rashid 1's wheels for a Material Adhesion and Abrasion Detection experiment.	link	verified-from-source
UAE10	Rashid 1 was carried by ispace's Hakuto-R Mission 1 lander, which failed to land on 2023-04-26 (JST); the rover never reached the surface.	link	verified-from-source
UAE11	ispace attributed the failure to software: the altitude sensor reading over a crater rim was discarded, so the lander believed it was at zero altitude when about 5 km up and ran out of propellant.	link	verified-from-source
UAE12	Rashid 1 landing target was Atlas crater (47.5N, 44.4E) on the southeastern edge of Mare Frigoris; UAE planned to change the site for the second rover.	link	verified-from-source

ID	Fact	Source	Status
UAE13	MBRSC's Director General said in May 2023 the team had not yet decided any technical improvements for the next rover, because Rashid 1 never operated on the Moon.	link	verified-from-source
UAE14	Rashid 2 will fly on Firefly Aerospace's Blue Ghost Mission 2 to the Moon's far side; Firefly said the rover will test various wheel materials for durability in lunar dust.	link	verified-from-source
UAE15	MBRSC completed Rashid 2 and shipped it to the US for preparation with Firefly; funded by the TDRA ICT Fund.	link	verified-from-source
UAE16	Rashid 2 testing included drive-off and deployment trials with Firefly in the US, antenna/arm deployment under simulated lunar gravity and vibration tests at EARTH (an EDGE subsidiary) in the UAE, and thermal-vacuum tes...	link	verified-from-source
UAE17	Rashid 2 launch timing sources conflict: Gulf News/Khaleej Times (Nov-Dec 2025) say 2026; Wikipedia Blue Ghost Mission 2 cites March 2026 reporting of NET Dec 2026 and infobox 2027; SpaceQ (2026-10-07) says Firefly list...	link	partially-verified
UAE18	Blue Ghost Mission 2 also carries Lunar Pathfinder, LuSEE-Night, Fleet Space SPIDER and other payloads; destination is the lunar far side.	link	partially-verified
UAE19	Hope (Emirates Mars Mission) is an orbiter, not a rover; launched from Tanegashima on an H-IIA on 19 July 2020 (UTC date varies by source) with Mars orbit insertion 9 Feb 2021.	link	verified-from-source
UAE20	Hope carries three instruments: a colour imager, infrared spectrometer and ultraviolet spectrometer, studying atmosphere and escape; no surface mobility role.	link	verified-from-source
UAE21	Mars Science City (announced Sept 2017): planned 177,000 sq m simulated Mars settlement, AED 500 million, domes of 3D-printed desert sand; the article does not mention rover or terrain testing.	link	verified-from-source
UAE22	TII (Abu Dhabi) presented an unmanned ground vehicle described as built to operate in desert without human intervention; named lead was Enrico Natalizio of TII's Autonomous Robotics Research Centre.	link	partially-verified
UAE23	xRift, an autonomous off-road vehicle by SteerAI (ATRC ecosystem, supported by SASC; TII engineers developed the base technology), debuted 20 Jan 2026 at UMEX for Gulf deserts and rocky terrain.	link	verified-from-source
UAE24	xRift specs reported: 4200×1880×1400 mm, 1,350 kg, 500 kg payload, 225 hp, max unmanned speed 50 km/h; production units expected by end of 2026.	link	verified-from-source
UAE25	Space42 (Abu Dhabi) showed a UGV at UMEX 2026 with 114 mm ground clearance, 25 degree slope capability and 10-80 km/h speed range; whether wheeled or tracked is not stated.	link	verified-from-source
UAE26	Emirates Robotics Competition (4th edition, RIT Dubai) task is a TIAGo Pro library book-retrieval challenge in the Mohammed Bin Rashid Library; Phase 1 in Gazebo simulation (deadline 2026-09-15), competition day Nov 202...	link	verified-from-source
UAE27	A recent arXiv scout-rover study on dune and lunar-simulant terrain (USC, UPenn, NASA) was tested at White Sands, New Mexico, not in the UAE.	link	verified-from-source

Appendix C. Sources (68 distinct URLs; each was retrieved on 2026-10-11)

#	Title	URL (retrieved 2026-10-11)	Evidence rows
1	Modeling of the interaction of rigid wheels with dry granular media (Agarwal et al., J. Terramechanics 2019 / arXiv 2019)	arxiv.org	TM-001, TM-002, TM-003, TM-004, TM-005, TM-006
2	General scaling relations for locomotion in granular media (Slonaker et al., arXiv 2016/PRE 2017)	arxiv.org	TM-007
3	Simulating Robotic Locomotion in Sand: Resistive Force Theory in an Open-Source Physics Engine (Brown, Treers, Daltorio; arXiv 2606.19504, under revi...	arxiv.org	TM-008, TM-009, TM-010
4	A resistive force model for legged locomotion on granular media (Li, Zhang, Goldman; arXiv 1911.00079)	arxiv.org	TM-011, TM-012
5	Granular Resistive Force Theory Implementation for Three-Dimensional Trajectories (Treers, Cao, Stuart; IEEE RA-L 2021)	edg.berkeley.edu	TM-013
6	Foot Shape-Dependent Resistive Force Model for Bipedal Walkers on Granular Terrains (arXiv 2403.03460)	arxiv.org	TM-014

#	Title	URL (retrieved 2026-10-11)	Evidence rows
7	Dynamic Wheel-Soil Model for Lightweight Mobile Robots with Smooth Wheels (Irani, Bauer, Warkentin; J. Intell. Robot. Syst. 2012)	carleton.ca	TM-015
8	Application of a Dynamic Pressure-Sinkage Relationship for Lightweight Mobile Robots (Irani, Bauer, Warkentin; Int. J. Veh. Auton. Syst., preprint)	carleton.ca	TM-016, TM-017
9	The effects of increasing velocity on the tractive performance of planetary rovers (Rodriguez-Martinez, Buse, Van Winnendael, Yoshida; arXiv 2306.021…	arxiv.org	TM-018, TM-019, LEG-022
10	Granular Scaling Laws for Accurate Prediction of Wheel Mobility on Slopes in Low-Gravity Environments (Omura, Ishigami; EasyChair preprint 14770 / IS…	easychair.org	TM-020
11	Modeling of slip rate-dependent traversability for path planning of wheeled mobile robot in sandy terrain (Sakayori, Ishigami; Frontiers in Robotics…	www.frontiersin.org	TM-021, LRN-021
12	Planetary rover mobility on loose soil: terramechanics theory for side slip prediction and compensation (Carletti, MSc thesis, Politecnico di Milano…	www.politesi.polimi.it	TM-022, TM-023, TM-024
13	Experimental Device for Measuring Sandy Soil Sinkage Parameters (Nguyen Van, Matsuo, Koumoto, Inaba; Bull. Fac. Agr. Saga Univ. 2007)	saga-u.repo.nii.ac.jp	TM-029
14	Using physics-based simulation towards eliminating empiricism in extraterrestrial terramechanics applications (Hu et al. 2024)	arxiv.org	SIM-001, SIM-002, SIM-003, SIM-004
15	Using high-fidelity discrete element simulation to calibrate an expeditious terramechanics model in a multibody dynamics framework (Zhang et al. 2024)	arxiv.org	SIM-005, SIM-006, SIM-007, SIM-008
16	A Physics-Based Continuum Model for Versatile, Scalable, and Fast Terramechanics Simulation (Unjhawala et al. 2025)	arxiv.org	SIM-009, SIM-010, SIM-011
17	Data-Driven Terramechanics Approach Towards a Realistic Real-Time Simulator for Lunar Rovers (Kern et al. 2026)	arxiv.org	SIM-012, SIM-013, SIM-014
18	Traction Performance Evaluation for a Rashid-1 Rover Wheel (Hurrell et al. 2025, Space Science Reviews 221:37)	doi.org	SIM-015, SIM-016, SIM-017
19	Soil Flow Analysis for Planetary Rovers Based on Particle Image Velocimetry and Discrete Element Method (Ono, Lichtenheldt, Yoshida; ISTVS 2020)	elib.dlr.de	SIM-018
20	DEM Analysis of Contact Forces and Tractive Performance of Rigid Wheel in Granular Media (Jelinek et al.; ISTVS 2021 preprint)	www.cavs.msstate.edu	SIM-019, SIM-020
21	Sinkage Study in Granular Material for Space Exploration Legged Robot Gripper (Candalot, Hurrell et al.; ISTVS 2024 paper 6422)	arxiv.org	SIM-021, SIM-022, SIM-023, SIM-024
22	RoverDevKit: An open, physics-grounded tradespace toolkit for conceptual design of lunar micro-rovers (Reifschneider 2026)	arxiv.org	SIM-025, SIM-026, SIM-027, SIM-028
23	Scout-Rover cooperation: online terrain strength mapping and traversal risk estimation for planetary-analog explorations (Liu et al. 2026)	arxiv.org	SIM-029
24	The Archimede Rover: A Comparison between Simulations and Experiments (Caruso et al., Robotics 2023, 12(5), 125)	doi.org	SIM-030
25	SlipNet: Enhancing Slip Cost Mapping for Autonomous Navigation on Heterogeneous and Deformable Terrains	arxiv.org	LRN-001, LRN-002
26	Slip Prediction Using Visual Information	m.roboticsproceedings.org	LRN-003, LRN-004, LRN-005
27	Terrain Adaptive Navigation for Planetary Rovers	www-robotics.jpl.nasa.gov	LRN-006
28	DeepTerramechanics: Terrain Classification and Slip Estimation for Ground Robots via Deep Learning	arxiv.org	LRN-007, LRN-008
29	Slippage prediction for off-road mobile robots via machine learning regression and proprioceptive sensing	hal.univ-grenoble-alpes.fr	LRN-009
30	Learning to Drive on Mars: Visual Multimodal Traversability Estimation for Off-World Navigation	arxiv.org	LRN-010, LRN-011
31	AI4MARS: A Dataset for Terrain-Aware Autonomous Driving on Mars	openaccess.thecvf.com	LRN-012
32	NeSAM: Neuro-Symbolic Kinodynamics with Soil Adaptation for Off-Road Mobility	arxiv.org	LRN-013, LRN-014
33	PrePARE: Predictive Proprioception for Agile Failure Event Detection in Robotic Exploration of Extreme Terrains	arxiv.org	LRN-015
34	Adaptive Meta-Learning for Identification of Rover-Terrain Dynamics	arxiv.org	LRN-016
35	Predicting Terrain Mechanical Properties in Sight for Planetary Rovers with Semantic Clues	arxiv.org	LRN-017
36	Physical Reservoir Computing in Hook-Shaped Rover Wheel Spokes for Real-Time Terrain Identification	arxiv.org	LRN-018

#	Title	URL (retrieved 2026-10-11)	Evidence rows
37	Learning All-Terrain Locomotion for a Planetary Rover with Actively Articulated Suspension	arxiv.org	LRN-019
38	Slip estimation model for traversability-based motion planning of cargo rover on extraterrestrial surface	www.frontiersin.org	LRN-020
39	Effect of Gait Design on Proprioceptive Sensing of Terrain Properties in a Quadrupedal Robot	arxiv.org	LRN-022
40	Proprioceptive Slip Detection for Planetary Rovers in Perceptually Degraded Extraterrestrial Environments	arxiv.org	LRN-024
41	Two Years of Visual Odometry on the Mars Exploration Rovers (Maimone, Cheng, Matthies; J. Field Robotics 2007)	robotics.jpl.nasa.gov	FLD-001, FLD-002, FLD-003, FLD-009
42	Characterization of traverse slippage experienced by Spirit rover on Husband Hill at Gusev crater (Li et al., JGR 2008)	authors.library.caltech.edu	FLD-004, FLD-014, FLD-015, FLD-016, FLD-017
43	Slip Sliding Away On The Red Planet (JPL news release reposted by Mars Daily, 2004-03)	www.marsdaily.com	FLD-005, FLD-006, FLD-007, FLD-008
44	Mars Exploration Rover Surface Operations: Driving Spirit at Gusev Crater (Leger et al., 2005)	www-robotics.jpl.nasa.gov	FLD-010, FLD-011, FLD-012, FLD-013
45	Curiosity Rover Mobility Issues: Crossing Martian Megaripple Fields (Arvidson, Maimone, Gildner, Hines; LPSC 2016 poster abstract 1137)	www-robotics.jpl.nasa.gov	FLD-018
46	Traction Control Design and Integration Onboard the Mars Science Laboratory Curiosity Rover (Toupet et al., IEEE Aerospace 2018)	www-robotics.jpl.nasa.gov	FLD-019, FLD-020, FLD-022
47	Proprioceptive Slip Detection for Planetary Rovers in Perceptually Degraded Extraterrestrial Environments (Kilic, Gu, Gross; arXiv 2207.13629)	arxiv.org	FLD-021
48	Roving on the Edge: Robotic Operations Power Perseverance's Ascent of Jezero Crater Rim (Maimone et al., AAS 08-138)	robotics.jpl.nasa.gov	FLD-023, FLD-024, FLD-025
49	Driving Strategies Used During Perseverance Rapid Traverse Campaign (Rankin et al., 2023)	robotics.jpl.nasa.gov	FLD-026
50	Analysis of Angle of Attack for Efficient Slope Ascent by Rovers (Inotsume, CMU-RI-TR-15-22 MS thesis)	publications.ri.cmu.edu	FLD-027, FLD-028, FLD-029, FLD-030, FLD-031
51	The Effect of Assistive Anchor-Like Grousers on Wheeled Rover Performance over Unconsolidated Sandy Dune Inclines (Sensors 2016, 16(9):1507)	www.ncbi.nlm.nih.gov	FLD-032, FLD-033, FLD-034, FLD-035
52	Traversing Steep and Granular Martian Analog Slopes With a Dynamic Quadrupedal Robot (Kolvenbach et al., arXiv 2106.01974)	arxiv.org	FLD-036
53	Failure Mechanisms and Risk Estimation for Legged Robot Locomotion on Granular Slopes (Liao and Qian, arXiv 2603.06928)	arxiv.org	FLD-037, FLD-038
54	Rovers in the desert (Emily Lakdawalla, The Planetary Society, 2012-05-14)	www.planetary.org	FLD-039
55	The Need for and Feasibility of Alternative Ground Robots to Traverse Sandy and Rocky Extraterrestrial Terrain (Li and Lewis, Adv. Intell. Syst. 2022…	arxiv.org	FLD-040
56	Self-driving rovers tested in Mars-like Morocco (ESA, 2018-12-19)	esa.int	FLD-041
57	AMADEE-18 Mission Report V1.2 public version (Austrian Space Forum, Oman Dhofar Mars analog, Feb 2018)	oewf.org	FLD-042
58	The effect of limb kinematics on the speed of a legged robot on granular media	arxiv.org	LEG-001, LEG-002
59	Sensitive dependence of the motion of a legged robot on granular media	arxiv.org	LEG-003, LEG-004
60	Systematic study of the performance of small robots on controlled laboratory substrates	goldmanlab.gatech.edu	LEG-005, LEG-006
61	Failure Mechanisms and Risk Estimation for Legged Robot Locomotion on Granular Slopes	arxiv.org	LEG-007, LEG-008, LEG-009
62	Traversing Steep and Granular Martian Analog Slopes With a Dynamic Quadrupedal Robot	arxiv.org	LEG-010, LEG-011, LEG-012, LEG-013
63	Virtual Energy Management for Physical Energy Savings in a Legged Robot Hopping on Granular Media	www.frontiersin.org	LEG-014
64	Learning Terrain-Adaptive Humanoid Locomotion on Granular Terrain	arxiv.org	LEG-015
65	Evaluation of Locomotion Performance for a Wheel-Paddle Robot (doctoral thesis, Ritsumeikan Univ.)	ritsumei.repo.nii.ac.jp	LEG-016
66	Traction Performance of Wheel and Track for Soft-Soil Traversal (Yoshida, ICRA 2010 planetary rovers workshop slides)	ewh.ieee.org	LEG-017, LEG-018, LEG-019, LEG-020
67	A Grouser Spacing Equation for Determining Appropriate Geometry of Planetary Rover Wheels	publications.ri.cmu.edu	LEG-021
68	OriWheelBot: An origami-wheeled robot	arxiv.org	LEG-023, LEG-024